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DENTAL PULP STEM CELL POPULATION

Abstract

The present invention relates, for example, to a stem cell population derived from human deciduous dental pulp, the stem cell population being characterized in that 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive, and to a method for producing a stem cell population derived from human deciduous dental pulp, comprising the step of culturing cells isolated from the human deciduous dental pulp in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate (hPL). The present invention also relates to an agent for preventing or treating cerebral palsy, comprising dental pulp stem cells, wherein the prevention or treatment agent is administered to a subject with symptoms of cerebral palsy.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a bypass continuing application based on and claiming priority to international application PCT/JP2023/035858, filed on Oct. 2, 2023, claiming the benefit of the filing dates of PCT/JP2022/036892, filed on Oct. 3, 2022, and Japanese Appl. No. 2023-080181, filed on May 15, 2022, and to international application PCT/JP2023/035859, filed on Oct. 2, 2023, claiming the benefit of the filing dates of Japanese Appl. No. 2022-159434, filed on Oct. 3, 2022, PCT/JP2022/036892, filed on Oct. 3, 2022, and Japanese Appl. No. 2023-071438, filed on Apr. 25, 2023, the content of each of which is incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to a stem cell population derived from human deciduous dental pulp and a culture supernatant thereof, a pharmaceutical composition comprising the stem cell population, and a method for producing the same. The present invention also relates to an agent for preventing or treating cerebral palsy, etc.

BACKGROUND ART

[0003] Stem cells derived from human dental pulp (dental pulp stem cells) have been reported to have a therapeutic effect on spinal cord injury, cerebral palsy (perinatal hypoxic ischemic encephalopathy), lower limb ischemia, and the like in experiments using animal models. However, according to the previous reports, the cells were administered directly to the spinal cord immediately after spinal cord injury (Non Patent Literature 1) or administered directly to the brain several hours after brain injury in cerebral palsy models (Non Patent Literature 2). These approaches are not realistic administration methods when clinical application is taken into consideration. In other cerebral palsy models, the cells were intravenously administered 24 hours after brain injury, although improvement in motor function was observed more than 5 months later (Non Patent Literature 3). Thus, there is a demand for a treatment method that produces an improvement effect earlier. As for lower limb ischemia, the cells were administered to model animals with moderate symptoms immediately after vascular ligation (Non Patent Literature 4) or several hours thereafter (Non Patent Literature 5). These approaches are detached from the actual status of treatment of severe lower limb ischemia mainly targeted by stem cell therapy.

[0004] Fetal bovine serum (FBS) has been used in methods for isolating dental pulp stem cells from dental pulp tissues (Non Patent Literatures 1 to 6). However, the possibility cannot be completely excluded that a foreign serum component remains in administered preparations. Therefore, use of FBS is not favorable when clinical application is taken into consideration. In the case of using cells isolated from dental pulp tissues of patients themselves, the cells may be prepared using the autologous serum of the patients. However, problems are a burden on the patients as well as the increase in time and cost required for preparing the cells using the

autologous serum on an individual basis. For rendering treatment with dental pulp stem cells available to more patients, it is desired to produce allogeneic cells in a large amount and use the cells as an off-the-shelf product to the treatment.

[0005] Dental pulp stem cells isolated by the conventional methods using FBS have been analyzed for their cellular characteristics. For example, characteristics have been reported in which the cell population comprises 50% or more or 96.2% CD117 (c-kit)-positive cells (Patent Literature 1 and Non Patent Literature 7). In the preceding report, CD117-positive cells have been separated and reported to have multipotency as their characteristics (Non Patent Literature 8). In addition, the dental pulp stem cells have been reported to be CD325 (N-cadherin)-positive (Non Patent Literature 9) and to include both cells positive and negative for CD51 (integrin (V)) and CD49d (integrin (4)) (Non Patent Literatures 10 and 11).

[0006] There is a report stating that a multipotent stem cell population is obtained by treating a human dental pulp tissue with trypsin and then culturing the tissue in a serum-free medium containing a human platelet lysate (hPL). However, the obtained cell population is positive for both of hematopoietic stem cell markers CD34 and CD45 and is therefore considered as a heterogeneous cell population (Non Patent Literature 12).

[0007] Cerebral palsy is caused by various perinatal cerebral damages including neonatal hypoxic-ischemic encephalopathy (HIE) and leads to dyskinesia or neurological symptoms such as epilepsy at the chronic stage (chronic phase). There is a need for the development of a treatment method that is effective even after symptoms of cerebral palsy are confirmed at the chronic stage.

[0008] Non Patent Literature 13 has reported that symptoms of cerebral palsy are improved as a result of administering mesenchymal stem cells derived from umbilical cord blood to the subarachnoid spaces of 4- to 14-year-old cerebral palsy patients. However, there are still few cases where umbilical cord blood is preserved at birth. Particularly, it is difficult to use mesenchymal stem cells derived from autologous umbilical cord blood (autologous transplantation) at the chronic stage. Stem cells, also including diverse stem cells other than the stem cells derived from umbilical cord blood, differ largely in the ability to differentiate, secretory factors, and the like, depending on their origins. Therefore, the type of stem cells having an improving effect on cerebral palsy symptoms cannot be predicted.

CITATION LIST

Patent Literature

[0009] [Patent Literature 1] WO2014/141210A2 (JP6491606B)

Non Patent Literature

[0010] [Non Patent Literature 1] Sakai et al., J Clin Invest. 2012 January; 122 (1): 80-90. doi: 10.1172/JCI59251. [0011] [Non Patent Literature 2] Yamagata et al., Stroke. 2013 February; 44 (2): 551-4. doi: 10.1161/STROKEAHA.112.676759. [0012] [Non Patent Literature 3] Kitase et al., Stem Cells Dev. 2020 Jan. 15; 29 (2): 63-74. doi: 10.1089/scd.2019.0221. PMID: 31801412. [0013] [Non Patent Literature 4] Yong et al., Tissue Eng Regen Med. 2022 August; 19 (4): 861-870. doi: 10.1007/s13770-022-00452-6. [0014] [Non Patent Literature 5] Li et al., Stem Cells Int. 2021 Aug. 31; 2021:5585255. doi: 10.1155/2021/5585255. [0015] [Non Patent Literature 6] Miura et al., Proc Natl Acad Sci USA. 2003 May 13; 100 (10): 5807-12. doi: 10.1073/pnas.0937635100. [0016] [Non Patent Literature 7] Suchanek et al., Acta Medica (Hradec Kralove). 2010; 53 (2): 93-9. doi: 10.14712/18059694.2016.66. [0017] [Non Patent Literature 8] Carnevale et al., J Tissue Eng Regen Med. 2018 February; 12 (2): e774-e785. doi: 10.1002/term.2378. [0018] [Non Patent Literature 9] Takahashi et al., Arch Oral Biol. 2012 January; 57 (1): 44-51. doi: 10.1016/j.archoralbio.2011.07.013. [0019] [Non Patent Literature 10] Lee et al., Stem Cell Res Ther. 2020 Jun. 3; 11 (1): 210. doi: 10.1186/s13287-020-01714-7. [0020] [Non Patent Literature 11] Hata et al., Stem Cell Res Ther. 2015 Sep. 7; 6 (1): 162. doi: 10.1186/s13287-015-0156-4. [0021] [Non Patent Literature 12] Pilbauerova et al., Biomolecules. 2022 Aug. 8; 12 (8): 1091. doi: 10.3390/biom12081091. [0022] [Non Patent Literature 13] Amanat et al., Stem Cell Research &

SUMMARY OF INVENTION

Technical Problem

[0023] An object of the present invention is to provide safe and highly functional stem cells suitable for clinical application, and a method for producing the same. Another object of the present invention is to provide an agent for preventing or treating cerebral palsy.

Solution to Problem

[0024] The inventors have found that a stem cell population obtained by culturing cells isolated from human deciduous dental pulp in a medium containing a human platelet lysate (hPL) has a structural feature (surface marker expression) different from that of previously reported dental pulp stem cells. The inventors have further found that the stem cell population obtained by this method has a higher amount of production of a cytokine useful for tissue regeneration than that of human deciduous tooth-derived dental pulp stem cells obtained by conventional methods using FBS and is capable of exerting excellent tissue regeneration capacity.

[0025] The present inventors also found that dental pulp stem cells have a prophylactic or therapeutic effect on cerebral palsy and a promoting effect on neurogenesis in the brain.

[0026] The present invention is based on these findings and provides the following [1] to according to the first embodiment.

[1] A (non-transgenic) stem cell population derived from human dental pulp, wherein 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive, and wherein the human dental pulp is human deciduous dental pulp.

[2] The stem cell population according to [1], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a serum-free medium containing a human platelet lysate.

[3] The stem cell population according to [1], wherein 90% or more of the stem cell population is CD325-negative or CD51-positive.

[4] The stem cell population according to [1], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0027] 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0028] 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0029] 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[5] The stem cell population according to [1], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0030] 1) producing at least 0.1 ng of SCF (stem cell factor) in 48 hours per 1×10^6 cells, [0031] 2) producing at least 500 ng of BIGH3 (transforming growth factor-beta-induced) in 48 hours per 1×10^6 cells, [0032] 3) producing at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.

[6] The stem cell population according to [1], wherein 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive; and [0033] the stem cell population has ability to produce any of BDNF, NGF, ANGPTL4, SDF-1 or MCP-1 at two times or more higher than bone marrow-derived mesenchymal stem cells (BMMSC), or adipose tissue-derived mesenchymal stem cells (ATMSC).

[7] The stem cell population according to [1], wherein the amount of production of IL-6 is reduced compared to a stem cell population derived from human deciduous dental pulp prepared using a medium containing FBS.

[8] The stem cell population according to [1], wherein the amount of production of one or more cytokines selected from the group consisting of SCF, ANGPTL4, BIGH3, BDNF, NGF, VEGF, SDF-1, MCP-1, and angiopoietin-2 is enriched compared to a stem cell population derived from human deciduous dental pulp prepared using a medium containing FBS.

[9] The stem cell population according to [1], wherein the stem cell population is obtained by the method comprising the steps of: [0034] enzymatically isolating cells from the human dental pulp;

[0035] culturing the cells in a serum-free medium containing a human platelet lysate; and [0036] obtaining a colony-forming cell population as the stem cell population.

[10] The stem cell population according to [9], wherein the method further comprises the step of expanding the obtained stem cell population for 2 to 8 passages for expansion.

[11] A culture supernatant of the stem cell population according to [1], wherein the culture supernatant has at least one of the following characteristics 1) to 3): [0037] 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0038] 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0039] 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[12] A composition comprising the stem cell population according to [1] and a pharmaceutically acceptable carrier.

[13] The composition according to [12], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a serum-free medium containing a human platelet lysate.

[14] The composition according to [12], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0040] 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0041] 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0042] 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[15] The composition according to [12], wherein the composition has at least one of the following characteristics 1) to 3): [0043] 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0044] 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0045] 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[16] A method for producing a stem cell population derived from human dental pulp, the method comprising the steps of: [0046] enzymatically isolating cells from the human dental pulp; [0047] culturing the cells in a serum-free medium in the presence of a human platelet lysate; and [0048] obtaining a colony-forming cell population as the stem cell population, [0049] wherein the human dental pulp is human deciduous dental pulp.

[17] The method according to [16], wherein the medium is animal-free.

[18] The method according to [16], wherein 90% or more of the resulting stem cell population is CD117-negative.

[19] The method according to [16], further comprising the step of expanding the obtained stem cell population for 2 to 8 passages for expansion.

[0050] The present invention provides the following [1] to according to the second embodiment.

[1] A method for treating a disease in need of tissue regeneration, the method comprising administering a (non-transgenic) stem cell population derived from human dental pulp and/or a culture supernatant of the stem cell population to a subject in need thereof, the stem cell population being characterized in that 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive, wherein the human dental pulp is human deciduous dental pulp.

[2] The method according to [1], wherein 90% or more of the stem cell population is CD325-negative or CD51-positive; preferably, 90% or more of the stem cell population is CD325-negative and is CD49d-positive and/or CD51-positive; more preferably, 90%, 95%, 97%, 98% or 99% or more of the stem cell population is CD325-negative and is CD49d-positive and/or CD51-positive.

[3] The method according to [1], wherein the stem cell population is CD31-negative.

[4] The method according to [1], wherein the stem cell population is CD34-negative and CD45-negative.

[5] The method according to [1], wherein the stem cell population produces SCF (stem cell factor)

at a weight ratio of 1/10 or more relative to IL-6.

[6] The method according to [1], wherein the stem cell population produces SCF (stem cell factor) at a weight ratio of 1/10 to 1/3 relative to IL-6.

[7] The method according to [1], wherein the stem cell population produces at least 0.1 ng of SCF in 48 hours per 1×10^6 cells.

[8] The method according to [1], wherein the stem cell population produces ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6.

[9] The method according to [1], wherein the stem cell population produces ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 to 20 relative to IL-6.

[10] The method according to [1], wherein the stem cell population produces at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.

[11] The method according to [1], wherein the stem cell population produces BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[12] The method according to [1], wherein the stem cell population produces BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 to 1000 relative to IL-6.

[13] The method according to [1], wherein the stem cell population produces at least 500 ng of BIGH3 in 48 hours per 1×10^6 cells.

[14] The method according to [1], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate (hPL).

[15] The method according to [14], wherein the medium is a serum-free medium.

[16] The method according to [14], wherein the medium is animal-free.

[17] The method according to [1], wherein the disease in need of tissue regeneration is any disease selected from spinal cord injury (including chronic-phase spinal cord injury), ischemic disease, inflammatory disease, autoimmune disease, and intestinal disease.

[18] The method according to [1], wherein the culture supernatant of the stem cell population has at least one of the following characteristics 1) to 3): [0051] 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0052] 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0053] 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[19] The method according to [1], wherein the culture supernatant of the stem cell population has at least one of the following characteristics 1) to 3): [0054] 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 to 1/3 relative to IL-6, [0055] 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 to 20 relative to IL-6, [0056] 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 to 1000 relative to IL-6.

[20] The method according to [1], wherein the human dental pulp is dental pulp of a deciduous tooth within 48 hours after tooth extraction.

[0057] According to the third embodiment, the present invention relates to the following [1] to:

[1] A composition for use in the prevention or treatment of cerebral palsy, comprising dental pulp stem cells, wherein the composition is administered to a subject with symptoms of cerebral palsy.

[2] The composition for use according to [1], wherein the cerebral palsy is chronic stage cerebral palsy.

[3] The composition for use according to [2], wherein the chronic stage is from infancy onward.

[4] The composition for use according to [1], wherein the composition is administered in multiple doses (2 to 20 doses, preferably 2 to 10 doses, further preferably 2 to 5 doses) to the subject.

[5] The composition for use according to [1], wherein the dental pulp stem cells are human deciduous dental pulp stem cells (stem cells derived from human deciduous dental pulp), preferably human deciduous dental pulp stem cells not induced to differentiate (particularly, not induced to differentiate into neurons) and/or non-transgenic human deciduous dental pulp stem cells.

[6] The composition for use according to [1], wherein the cerebral palsy is cerebral palsy caused by

hypoxia and/or ischemia.

[7] The composition for use according to [1], wherein the cerebral palsy is caused by perinatal cerebral damage, such as Perinatal Hypoxic-Ischemic encephalopathy.

[8] The composition for use according to [1], wherein the composition is used for improving a motor function and/or a memory learning function of the subject with symptoms of cerebral palsy.

[9] The composition for use according to [1], wherein the subject is a human, for example, a human of 6 months or later after birth or a human who has undergone neonatal asphyxia and/or been diagnosed with periventricular leukomalacia.

[10] The composition for use according to [1], wherein the composition causes neurogenesis in the brain of the subject.

[11] The composition for use according to [1], wherein the dental pulp stem cells consist of a stem cell population derived from human deciduous dental pulp, the stem cell population being characterized in that 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive.

[12] The composition for use according to [11], wherein 90% or more of the stem cell population is CD325-negative or CD51-positive.

[13] The composition for use according to wherein the stem cell population has at least one of the following characteristics 1) to 3): [0058] 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0059] 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0060] 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[14] The composition for use according to [11], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0061] 1) producing at least 0.1 ng of SCF (stem cell factor) in 48 hours per 1×10^6 cells, [0062] 2) producing at least 500 ng of BIGH3 (transforming growth factor-beta-induced) in 48 hours per 1×10^6 cells, [0063] 3) producing at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.

[15] The composition for use according to [11], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate.

[0064] According to the fourth embodiment, the present invention relates to the following [1] to:

[1] A method for preventing or treating cerebral palsy, comprising administering dental pulp stem cells to a subject, wherein the subject is a subject with symptoms of cerebral palsy.

[2] The method according to [1], wherein the cerebral palsy is chronic stage cerebral palsy.

[3] The method according to [2], wherein the chronic stage is from infancy onward.

[4] The method according to [1], wherein the dental pulp stem cells are administered in multiple doses (2 to 20 doses, preferably 2 to 10 doses, further preferably 2 to 5 doses).

[5] The method according to [1], wherein the dental pulp stem cells are human deciduous dental pulp stem cells (stem cells derived from human deciduous dental pulp).

[6] The method according to [1], wherein the cerebral palsy is cerebral palsy caused by hypoxia and/or ischemia.

[7] The method according to [1], wherein the cerebral palsy is caused by perinatal cerebral damage, such as Perinatal Hypoxic-Ischemic encephalopathy.

[8] The method according to [1], wherein the dental pulp stem cells improve a motor function and/or a memory learning function of the subject with symptoms of cerebral palsy.

[9] The method according to [1], wherein the subject is a human.

[10] The method according to [1], wherein the dental pulp stem cells cause neurogenesis in the brain of the subject.

[11] The method according to [1], wherein the dental pulp stem cells consist of a stem cell population derived from human deciduous dental pulp, the stem cell population being characterized in that 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive,

and CD105-positive.

[12] The method according to [11], wherein 90% or more of the stem cell population is CD325-negative or CD51-positive.

[13] The method according to [11], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0065] 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0066] 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, [0067] 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[14] The method according to [11], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0068] 1) producing at least 0.1 ng of SCF (stem cell factor) in 48 hours per 1×10^6 cells, [0069] 2) producing at least 500 ng of BIGH3 (transforming growth factor-beta-induced) in 48 hours per 1×10^6 cells, [0070] 3) producing at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.

[15] The method according to [11], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate.

[16] The method according to [1], wherein the dental pulp stem cells are administered as a single active ingredient.

[17] The method according to [1], wherein the dental pulp stem cells are human deciduous dental pulp stem cells not induced to differentiate and/or non-transgenic human deciduous dental pulp stem cells.

[18] The method according to [1], wherein the dental pulp stem cells are human deciduous dental pulp stem cells not induced to differentiate into neurons.

[19] The method according to [1], wherein the subject is a human who has undergone neonatal asphyxia and/or been diagnosed with periventricular leukomalacia.

[20] The method according to [1], wherein the subject is a human of six months or later after birth.

[0071] According to the fifth embodiment, the present invention relates to the following [1] to:

[1] Use of dental pulp stem cells in the production of an agent for preventing or treating cerebral palsy, wherein the prevention or treatment agent is for a subject with symptoms of cerebral palsy.

[2] The use according to [1], wherein the cerebral palsy is chronic stage cerebral palsy.

[3] The use according to [2], wherein the chronic stage is from infancy onward.

[4] The use according to any of [1] to [3], wherein the dental pulp stem cells are human deciduous dental pulp stem cells (stem cells derived from human deciduous dental pulp).

[5] The use according to any of [1] to [4], wherein the cerebral palsy is cerebral palsy caused by hypoxia and/or ischemia.

[6] The use according to any of [1] to [5], wherein the cerebral palsy is caused by perinatal cerebral damage.

[7] The use according to any of [1] to [6], wherein the prevention or treatment agent is for improving a motor function and/or a memory learning function of the subject with symptoms of cerebral palsy.

[8] The use according to any of [1] to [7], wherein the subject is a human.

[9] The use according to any of [1] to [8], wherein the dental pulp stem cells consist of a stem cell population derived from human deciduous dental pulp, the stem cell population being characterized in that 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive.

[10] The use according to [9], wherein 90% or more of the stem cell population is CD325-negative or CD51-positive.

[11] The use according to [9] or [10], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0072] 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, [0073] 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio

of 5 or more relative to IL-6, [0074] 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

[12] The use according to any of [9] to [11], wherein the stem cell population has at least one of the following characteristics 1) to 3): [0075] 1) producing at least 0.1 ng of SCF (stem cell factor) in 48 hours per 1×10^6 cells, [0076] 2) producing at least 500 ng of BIGH3 (transforming growth factor-beta-induced) in 48 hours per 1×10^6 cells, [0077] 3) producing at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.

[13] The use according to any of [9] to [12], wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate.

Advantageous Effects of Invention

[0078] The present invention provides a highly functional and safe dental pulp stem cell population. Unlike conventional methods, the stem cell population of the present invention is prepared using a serum-free medium containing no Xenogeneic component such as FBS and obtained as a highly homogeneous cell population without any special separation approach. The present invention can also provide an agent for preventing or treating cerebral palsy using dental pulp stem cells.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0079] FIG. 1A and FIG. 1B show surface marker analysis results of the human dental pulp stem cells according to the present invention. FIG. 1A: CD31, FIG. 1B: CD34

[0080] FIG. 2 shows a microscopic image obtained by the oil red staining of adipocytes (fat droplets) induced by differentiation from the human dental pulp stem cells.

[0081] FIG. 3 shows a microscopic image obtained by the calcification staining of osteoblasts induced by differentiation from the human dental pulp stem cells.

[0082] FIG. 4 shows a microscopic image obtained by the alcian blue staining of a cartilage tissue induced by differentiation from the human dental pulp stem cells.

[0083] FIG. 5 shows immunostaining images of neurons induced by differentiation from the human dental pulp stem cells (isotype control, anti-Nestin, anti- β III-tubulin, anti-neurofilament M immunostaining images in order from the left).

[0084] FIG. 6 shows a neurite elongating effect of a culture supernatant of the human dental pulp stem cells.

[0085] FIG. 7 shows a neural progenitor cell proliferative effect of a culture supernatant of the human dental pulp stem cells.

[0086] FIG. 8 shows a vascular endothelial cell proliferative effect of a culture supernatant of the human dental pulp stem cells.

[0087] FIG. 9 shows a vascular-like structure constructing effect (tube formation) of a culture supernatant of the human dental pulp stem cells.

[0088] FIG. 10 shows a vascular endothelial cell recruiting effect of a culture supernatant of the human dental pulp stem cells.

[0089] FIG. 11 shows an immunosuppressive effect (ability to suppress the proliferation of CD4-positive T cells) of the human dental pulp stem cells.

[0090] FIG. 12A and FIG. 12B show a motor function improving effect (A) and nerve regenerative effect (B) of the human dental pulp stem cells on chronic-phase spinal cord injury in a rat spinal cord injury model.

[0091] FIG. 13 shows a motor function improving effect of the human dental pulp stem cells on perinatal hypoxic ischemic encephalopathy in a newborn rat hypoxic ischemic encephalopathy

model.

[0092] FIG. **14** shows an improving effect on the volume of femoral blood flow (A) and an improving effect on limb necrosis (B) in an immunodeficient rat severe lower limb ischemia model.

[0093] FIG. **15** shows an amount of BDNF production (A) and an amount of VEGF production (B) of the human dental pulp stem cells.

[0094] FIG. **16** shows a vascular endothelial cell proliferative effect of a culture supernatant of the human dental pulp stem cells.

[0095] FIG. **17** shows a vascular-like structure constructing effect (tube formation) of a culture supernatant of the human dental pulp stem cells.

[0096] FIG. **18** shows amounts of cytokine production of various stem cells.

[0097] FIG. **19** shows an increasing effect on a vessel area (A) and an increasing effect on a vessel number (B) in an immunodeficient rat severe lower limb ischemia model.

[0098] FIG. **20** shows anti-inflammatory effects of various stem cells under M2 induction conditions (A) and M1 induction conditions (B) of THP-1 cells.

[0099] FIG. **21** shows proliferation rates of human dental pulp stem cell culture in various serum-free media and a serum-free medium containing hPL.

[0100] FIG. **22** shows a motor function improving effect of the human dental pulp stem cells on severe perinatal hypoxic ischemic encephalopathy in a newborn rat severe perinatal hypoxic ischemic encephalopathy model.

[0101] FIG. **23** shows results of a horizontal ladder test in the case of single-dose administration of cells (Test Example 2).

[0102] FIG. **24** shows results of a horizontal ladder test in the case of multiple-dose administration of cells (Test Example 2).

[0103] FIG. **25** shows a diagram in which the results of the horizontal ladder test (Test Example 2) were compared using the single-dose administration of cells and the multiple-dose administration of cells.

[0104] FIG. **26** shows results of a cylinder test in the case of multiple-dose administration of cells (Test Example 2).

[0105] FIG. **27** shows results of shuttle avoidance in the case of multiple-dose administration of cells (Test Example 2).

[0106] FIG. **28** shows results of measuring a body weight (Test Example 3).

[0107] FIG. **29** shows results of measuring a brain weight ratio (Test Example 3).

[0108] FIG. **30** shows a fluorescence intensity ratio of deciduous dental pulp stem cells after administration of the cells (which indicates the abundance of the cells).

[0109] FIG. **31** shows the ratio of BrdU/DCX double stain-positive cells in a hippocampal dentate gyrus lesion of the brain after administration of the cells.

[0110] FIG. **32** shows the number of BrdU-positive cells in a hippocampal dentate gyrus lesion of the brain after administration of the cells.

[0111] FIG. **33** shows the number of NeuN-positive cells in a hippocampal dentate gyrus lesion of the brain after administration of the cells.

[0112] FIG. **34** shows results of a rotarod test of a perinatal hypoxic-ischemic encephalopathy model in the case of single-dose administration of cells (Test Example 7).

[0113] FIG. **35** shows results of a cylinder test of a severe perinatal hypoxic-ischemic encephalopathy model in the case of multiple-dose administration of cells (Test Example 8).

[0114] FIG. **36** shows a neural stem cell proliferation rate by the coculture of neural stem cells and deciduous dental pulp stem cells, bone marrow-derived mesenchymal stem cells, or fibroblasts (Test Example 9).

[0115] FIG. **37** shows the amount of BDNF in a culture supernatant by coculture with neural stem cells (Test Example 9).

DESCRIPTION OF EMBODIMENTS

[0116] The terms “contain (containing)” and “comprise (comprising)” as used herein conceptually include the terms “contain (containing)”, “comprise (comprising)”, “consist (consisting) essentially of”, and “consist (consisting) only of”.

1. Stem Cell Population Derived from Human Dental Pulp

[0117] The present invention relates to a non-transgenic stem cell population derived from human dental pulp, at least 90% or more of which is CD117-negative, CD73-positive, CD90-positive, and CD105-positive (hereinafter, also referred to as the “stem cell population of the present invention”). Hereinafter, structural or functional characteristics of the stem cell population of the present invention will be described.

1.1 Origin

[0118] The stem cell population of the present invention is a non-transgenic stem cell population derived from human dental pulp. The “dental pulp” may be any dental pulp of a deciduous tooth or a permanent tooth, and an extracted tooth such as a deciduous tooth or a wisdom tooth is preferred because of its easy availability, etc. It is also desired to use a tooth within 72 hours after tooth extraction, more preferably within 48 hours after tooth extraction. It is particularly desired to use a human deciduous tooth preferably within 48 hours after tooth extraction.

[0119] The stem cell population of the present invention may be autologous cells derived from dental pulp of a recipient or may be allogeneic cells derived from dental pulp of another person. The stem cell population is preferably allogeneic cells from the viewpoint of time and cost required for provision and stable quality of the cells.

1.2 Marker

[0120] “CD117” is a transmembrane protein that functions as a tyrosine kinase receptor and is also called c-kit. Previously known dental pulp stem cells are CD117-positive or are a heterogeneous population comprising CD117-positive cells and CD117-negative cells (Hilken et al., *Cell and Tissue Research* (2013) 353, 65-78; Deng et al., *Frontiers in Cell and Developmental Biology*, March 2021, volume 9, Article 661116; Ferro et al., *PLOS ONE* July 2012, volume 7, Issue 7, e41774; and Lei et al., *Stem Cell International*, Volume 2021, Article ID 8886854). By contrast, the stem cell population of the present invention is characterized by being CD117-negative. Also, the stem cell population of the present invention is characterized by being positive for mesenchymal stem cell markers CD73, CD90, and CD105.

[0121] “CD325”, “CD49d”, and “CD51” are cell surface proteins that function as adhesion factors and are also called N-cadherin (cadherin 2), integrin 04, and integrin CV, respectively. Previously known dental pulp stem cells are CD325-positive or are a heterologous population comprising CD325-positive cells and CD325-negative cells (Deng et al., *Frontiers in Cell and Developmental Biology*, March 2021, volume 9, Article 661116; and Madhoun et al., *Frontiers in Cell and Developmental Biology*, October 2021, volume 9, Article 717624). By contrast, the stem cell population of the present invention is preferably CD325-negative. Previously known dental pulp stem cells are a heterologous population comprising both cells positive and negative for CD51 and CD49d (Lei et al., *Stem Cell International*, Volume 2021, Article ID 8886854; and Alvarez et al., *International Journal of Oral Science* (2015), 7, 205-212). By contrast, the stem cell population of the present invention preferably exhibits a CD49d- and/or CD51-positive ratio of 90%, 95%, 97%, 98%, or 99% or more.

[0122] The stem cell population of the present invention is preferably negative for an endothelial cell marker CD31 and also negative for hematopoietic stem cell markers such as CD34-negative and CD45-negative. The stem cell population of the present invention has a positive ratio of 90% or more, preferably 95% or more, for CD150, a positive marker of mesenchymal stem cells, and is preferably negative for negative markers CD14 and CD19.

[0123] The stem cell population of the present invention is preferably negative for HLA-DR, HLA-DQ, CD40, CD80, and CD86 related to immunogenicity.

[0124] The stem cell population of the present invention is preferably positive for adhesion factors CD29 (ITGB1), CD44, and CD166 (ALCAM) and negative for CD106 (VCAM).

[0125] The stem cell population of the present invention is CD73-positive, CD90-positive, and CD105-positive, preferably is CD73-positive, CD90-positive, and CD105-positive, and is CD117-negative, further preferably is CD73-positive, CD90-positive, and CD105-positive, and is CD117-negative and CD325-negative.

[0126] The term “positive” for a marker protein or a marker gene, as used herein, means that the stem cell population exhibits a positive ratio of 30% or more, preferably 40% or more, 50% or more, 60% or more, 70% or more, or 80% or more, more preferably 90% or more, for the protein or the gene when the stem cell population at passages 2 to 8 is assayed by an approach known in the art (e.g., detection data from flow cytometry is analyzed with an isotype control). Further preferably, the term “positive” for a marker protein or a marker gene means that the stem cell population at passages 2 to 8 has a positive ratio of 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% or more for the protein or the gene. Particularly preferably, the term “positive” to a marker protein or a marker gene means that the stem cell population at passages 2 to 8 has a positive ratio of 99.1% or more, 99.2% or more, 99.3% or more, 99.4% or more, 99.5% or more, 99.6% or more, 99.7% or more, 99.8% or more, 99.9% or more, or 100% to the protein or the gene. The term “negative” for a marker protein or a marker gene means that the stem cell population has a positive ratio of less than 10%, preferably less than 7%, more preferably less than 6% or less than 5%, further preferably less than 3% or less than 2%, for the protein or the gene when detected in the same manner as above (however, the value of negativity does not fall below 1% when fluorescence intensity of 1% in an isotype control is set as a positive-negative threshold). In this context, the passage of the cells is performed in a 60% confluent state for adhesion culture.

[0127] The cell population of the present invention has the following features: [0128] (1) having a CD117-positive ratio of preferably less than 3%, more preferably less than 2%; [0129] (2) having a CD73-positive ratio of preferably 98% or more, more preferably 99% or more; [0130] (3) having a CD90-positive ratio of preferably 98% or more, more preferably 99% or more; and [0131] (4) having a CD105-positive ratio of preferably 98% or more, more preferably 99% or more at passages 2 to 8.

[0132] The cell population of the present invention may further have, in addition to the features (1) to (4), any two or more, three or more, or four or more of the following features (5) to (20) at passages 2 to 8: [0133] (5) having a CD325-positive ratio of preferably less than 7%, more preferably less than 5%, [0134] (6) having a D49d- and/or CD51-positive ratio of preferably 97% or more, more preferably 98% or more, further preferably 99% or more, [0135] (7) having a CD31-positive ratio of preferably less than 3%, more preferably less than 2%, [0136] (8) having a CD34-positive ratio of preferably less than 7%, more preferably less than 5%, [0137] (9) having a CD45-positive ratio of preferably less than 7%, more preferably less than 5%, [0138] (10) having a CD150-positive ratio of preferably 90% or more, more preferably 95% or more, [0139] (11) having a CD14-positive ratio of preferably less than 7%, more preferably less than 5%, [0140] (12) having a CD19-positive ratio of preferably less than 7%, more preferably less than 5%, [0141] (13) having HLA-DR- and HLA-DQ-positive ratios each of preferably less than 7%, more preferably less than 5%, [0142] (14) having a CD40-positive ratio of preferably less than 7%, more preferably less than 5%, [0143] (15) having a CD80-positive ratio of preferably less than 7%, more preferably less than 5%, [0144] (16) having a CD86-positive ratio of preferably less than 7%, more preferably less than 5%, [0145] (17) having a CD29-positive ratio of preferably 96% or more, more preferably 98% or more, [0146] (18) having a CD44-positive ratio of preferably 96% or more, more preferably 98% or more, [0147] (19) having a CD166-positive ratio of preferably 96% or more, more preferably 98% or more, [0148] (20) having a CD106-positive ratio of preferably less than 7%, more preferably less than 5%.

[0149] The detection of a marker protein can be carried out by immunological assay using an

antibody, such as ELISA, immunostaining, or flow cytometry. In the case of a protein that is intracellularly expressed without appearing on cell surface, the protein of interest can be detected by expressing a reporter protein together with the protein and detecting the reporter protein. The detection of a marker gene can be carried out by use of a nucleic acid amplification method and/or a nucleic acid detection method, such as RT-PCR, a microarray, or a biochip.

1.3 Cytokine Production

[0150] The stem cell population of the present invention has a high amount of production of a cytokine useful for tissue regeneration and is rich in the cytokines, as compared with conventional dental pulp stem cells prepared using a medium containing FBS. Particularly, the stem cell population of the present invention has the feature that the ratio of the amount of production of a cytokine useful for tissue regeneration to the amount of production of an inflammatory cytokine is high as compared with the conventional dental pulp stem cells.

[0151] Examples of the cytokine useful for tissue regeneration can include SCF (stem cell factor), ANGPTL4 (angiopoietin-like 4), BIGH3 (transforming growth factor-beta-induced), brain-derived neurotrophic factor (BDNF), nerve growth factor (NGF), vascular endothelial growth factor (VEGF), stromal cell-derived factor 1 (SDF-1), monocyte chemoattractant protein-1 (MCP-1), and angiopoietin-2. Examples of the inflammatory cytokine can include IL-6 (interleukin-6).

[0152] “SCF” (stem cell factor) is a protein that is generated and secreted (in the present specification, collectively referred to as “produced”) by mammalian cells, and is known as a nutritional factor that allows hematopoietic stem cells to proliferation in cultured cell experiments. Also, SCF has been reported to act on neurons or vascular endothelial cells expressing its receptor CD117 and thereby promote neuroprotection (Dhandapani et al., *J Neurochem.* 2005 October; 95 (1): 9-19), neurogenesis (Jin et al., *J Clin Invest.* 2002 August; 110 (3): 311-9; and Osada et al., *J Neurosurg Spine.* 2010 October; 13 (4): 516-23), neurite elongation (Hirata et al., *Development.* 1993 September; 119 (1): 49-56), and angiogenesis (Matsui et al., *J Biol Chem.* 2004 Apr. 30; 279 (18): 18600-7; Sun L., *Cancer Cell.* 2006 April; 9 (4): 287-300; and Kim et al., *Cardiovasc Res.* 2011 Oct. 1; 92 (1): 132-40) in cultured cell experiments or animal experiments. These effects are useful in the treatment of not only nerve disease and ischemic disease but various diseases in need of tissue regeneration, such as intestinal disease and bone disease.

[0153] “ANGPTL4” (angiopoietin-like 4) is a protein that is generated and secreted by mammalian cells, and has been reported to have an angiogenic effect (Le Jan et al., *Am J Pathol.* 2003 May; 162 (5): 1521-8; Hermann et al., *Clin Immunol.* 2005 April; 115 (1): 93-101; and Ma et al., *Proc Natl Acad Sci USA.* 2010 Aug. 10; 107 (32): 14363-8) and an anti-inflammatory effect (Cho et al., *JCI Insight.* 2019 Aug. 22; 4 (16): e125437) in cultured cell experiments or animal experiments. These effects are useful in the treatment of not only ischemic disease, inflammatory disease, and autoimmune disease but also various diseases in need of tissue regeneration.

[0154] “BIGH3” (transforming growth factor-beta-induced) is a protein that is generated and secreted by mammalian cells, and is also called β ig-H3, keratoepithelin, or TGFBI. BIGH3 has been reported to have an angiogenic effect (Aitkenhead et al., *Microvasc Res.* 2002 March; 63 (2): 159-71) and a bone or cartilage degradation suppressive effect (Ruiz et al., *Biomaterials.* 2020 January; 226:119544) in cultured cell experiments or animal experiments. These effects are considered useful in the treatment of not only ischemic disease and bone or cartilage disease but also various diseases in need of tissue regeneration.

[0155] “BDNF” (brain-derived neurotrophic factor) is a protein that is generated and secreted by mammalian cells, and is widely distributed in the brain of mammals. BDNF, which is a neurotrophic factor, is known to promote the development and elongation of neurons and participate in the regulation of synaptic functions (Kowianski et al., *Cell Mol Neurobiol* (2018) 38:579-593), and has been reported to have a neuroprotective effect (Lau et al., *Cell Reports* (2015) 12:1353-1366; and Numakawa et al., *Histol Histopathol* (2010) 25:237-258). These effects are considered useful in the treatment of not only neurodegenerative disease but various diseases in

of nerve tissue regeneration.

[0156] “VEGF” (vascular endothelial growth factor) is a protein that is generated and secreted by mammalian cells. VEGF is known to induce the cell division, migration, and differentiation of vascular endothelial cells and thereby cause angiogenesis. VEGF is also considered to participate in angiogenesis or neurogenesis at injury sites through the induction of its expression in nerve tissues that have gotten a hypoxic state due to spinal cord injury (Long et al., Chinese Journal of Traumatology 18 (2015) 293-295). In addition, VEGF is also considered to contribute to axonal regeneration through angiogenesis at nerve injury sites in the peripheral nervous system (Saio et al., Int. J. of Mol. Sci. 2021, 22, 11169). These effects are considered useful in the treatment of not only neurodegenerative disease but also various diseases in need of nerve tissue regeneration.

[0157] “NGF” (nerve growth factor) is a protein that is generated and secreted by mammalian cells, and is known to have a neuroprotective effect in the central nervous system and to participate in the development or survival of neurons in the peripheral nervous system (Keefe et al., Int. J. of Mol. Sci. 2017, 18, 548; and Mariga et al., Neurobiol Dis 2017 January; 97 (Pt B): 73-79). These effects are considered useful in the treatment of not only neurodegenerative disease but also various diseases in need of nerve tissue regeneration.

[0158] “MCP-1” (monocyte chemoattractant protein-1) is a protein that is generated and secreted by mammalian cells, and is also called C—C motif chemokine 2 (CCL2). MCP-1 is known to have the effects to enhance chemotaxis of monocyte and T lymphocyte and the ability to induce the differentiation of dendritic cells, and to participate in, for example, neurodegenerative disease (Parkinson's disease, Alzheimer's disease, multiple sclerosis, etc.) whose onset is triggered by chronic inflammation (Singh et al., International Immunopharmacology 101 (2021) 107598). Also, MCP-1 has been reported to restore the volume of blood flow by improving collateral circulation via arteriogenesis (Ito et al., Circulation Res. (1997), vol. 80, Issue 6, 829-837). Thus, MCP-1 is considered useful in the treatment of inflammatory disease, neurodegenerative disease, and various diseases in need of vascular formation or nerve tissue regeneration.

[0159] “Angiopoietin-2” is a protein that is generated and secreted by mammalian cells, and is known to participate in lymphangiogenesis in lymphatic endothelial cells and vascular remodeling in tissues (Akwii et al., Cells 2019, 8, 471). Angiopoietin-2 is considered useful in the treatment of diseases in need of angiogenesis and tissue regeneration.

[0160] “IL-6” is a protein that is generated and secreted by mammalian cells, and is known as an inflammatory cytokine that increases immune response and tissue inflammation. Also, IL-6 is a representative factor of a series of inflammatory cytokine groups called SASP (senescence-associated secretory phenotype) produced by senile cells, and is considered responsible for chronic inflammation associated with senescence (Rolt et al., Biogerontology. 2019 June; 20 (3): 359-371; and Di et al., PLOS One. 2014 Nov. 24; 9 (11): e113572). Thus, it is not preferred that cells administered for the purpose of treating a disease should produce a large amount of IL-6.

[0161] Specifically, the stem cell population of the present invention is capable of producing SCF at a weight ratio of at least 1/20 or more, preferably a weight ratio of at least 1/10 or more, more preferably a weight ratio of 1/10 to 1/2, further preferably a weight ratio of 1/10 to 1/3, more preferably a weight ratio of at least 1/5 or more, further preferably a weight ratio of at least 1/3 or more, relative to IL-6 (however, without exceeding the amount of production of IL-6) at passages 2 to 8. The stem cell population of the present invention is capable of producing at least 0.1 ng, 0.13 ng, or 0.15 ng, preferably at least 0.17 ng, more preferably at least 0.20 ng, of SCF in 48 hours of culture per 1×10^6 cells at passages 2 to 8.

[0162] As in previously known dental pulp stem cells, when cells themselves to be administered to a subject are CD117-positive, SCF produced by the administered cells are first taken up into the administered cells themselves via CD117 of the self or other administered cells. This reduces its action efficiency on endogenous cells of a recipient on which SCF is supposed to act. The stem cell population of the present invention has high ability to produce SCF and is CD117-negative, and as

such, is capable of exerting an excellent tissue regenerating effect when administered to a subject in need of tissue regeneration.

[0163] Specifically, the stem cell population of the present invention has a CD117-positive ratio of preferably less than 3%, more preferably less than 2%; and is capable of producing SCF at a weight ratio of at least 1/20 or more, preferably a weight ratio of at least 1/10 or more, more preferably a weight ratio of at least 1/5 or more, further preferably a weight ratio of at least 1/3 or more, relative to IL-6 (however, without exceeding the amount of production of IL-6) at passages 2 to 8.

[0164] The stem cell population of the present invention is capable of producing ANGPTL4 at a weight ratio of at least 3 or more, preferably a weight ratio of at least 5 or more, more preferably a weight ratio of 5 to 20, further preferably a weight ratio of 5 to 15, further preferably a weight ratio of at least 10 or more, particularly preferably a weight ratio of at least 20 or more, relative to IL-6 at passages 2 to 8. The stem cell population of the present invention is capable of producing at least 4 ng, 4.5 ng, or 5.0 ng, preferably at least 6.9 ng, more preferably at least 7.0 ng, of ANGPTL4 in 48 hours of culture per 1×10^6 cells at passages 2 to 8.

[0165] The stem cell population of the present invention is capable of producing BIGH3 at a weight ratio of 400 or more, preferably a weight ratio of 450 or more, more preferably a weight ratio of 450 to 1500, further preferably a weight ratio of 450 to 1000, relative to IL-6 at passages 2 to 8. The stem cell population of the present invention is capable of producing at least 400 ng, 450 ng, or 500 ng, preferably at least 550 ng, more preferably at least 600 ng, of BIGH3 in 48 hours per 1×10^6 cells at passages 2 to 8.

[0166] In this context, the weight ratio of each liquid factor in a culture supernatant of the stem cell population of the present invention is determined by measuring the amount of each liquid factor contained in the culture supernatant after culture.

[0167] The stem cell population of the present invention is a cell population capable of producing SCF at a weight ratio of at least 1/20 or more, preferably a weight ratio of at least 1/10 or more, more preferably a weight ratio of at least 1/5 or more, further preferably a weight ratio of at least 1/3 or more, relative to IL-6 (however, without exceeding the amount of production of IL-6); producing ANGPTL4 at a weight ratio of at least 3 or more, preferably a weight ratio of at least 5 or more, more preferably a weight ratio of at least 10 or more, further preferably a weight ratio of at least 20 or more, relative to IL-6; and producing BIGH3 at a weight ratio of 400 or more, preferably a weight ratio of 450 or more, more preferably a weight ratio of 450 to 1500, further preferably a weight ratio of 450 to 1000, relative to IL-6 at passages 2 to 8.

[0168] The stem cell population of the present invention is capable of producing SCF at a weight ratio of at least 1/10 or more, preferably a weight ratio of at least 1/5 or more, more preferably a weight ratio of at least 1/2 or more, further preferably at a weight ratio of at least 1/3 or more, relative to IL-6 (however, without exceeding the weight of production of IL-6) at passages 2 to 8 when cultured in the presence of a human platelet lysate (hPL) in a medium free of other animal-derived components (particularly, FBS). The stem cell population of the present invention is capable of producing SCF at a weight ratio of at least approximately 3 or more, preferably at least approximately 5 or more, further preferably at least approximately 7 or more when cultured in the presence of hPL in a medium free of other animal-derived components, as compared with when cultured in a medium containing FBS.

[0169] The stem cell population of the present invention is capable of producing ANGPTL4 at a weight ratio of at least approximately 5 or more, preferably a weight ratio of at least approximately 10 or more, further preferably a weight ratio of at least approximately 20 or more, relative to IL-6 at passages 2 to 8 when cultured in the presence of a human platelet lysate (hPL) in a medium free of other animal-derived components (particularly, FBS). The stem cell population of the present invention is capable of producing ANGPTL4 at a weight ratio of at least approximately 3 or more, preferably at least approximately 5 or more, further preferably at least approximately 7 or more, when cultured in the presence of hPL in a medium free of other animal-derived components, as

compared with when cultured in a medium containing FBS.

[0170] The stem cell population of the present invention is capable of producing VEGF at a weight ratio of at least approximately 2 or more, preferably a weight ratio of at least approximately 5 or more, further preferably a weight ratio of at least approximately 10 or more, relative to IL-6 at passages 2 to 8 when cultured in the presence of a human platelet lysate (hPL) in a medium free of other animal-derived components (particularly, FBS). The stem cell population of the present invention is capable of producing VEGF at a weight ratio of at least approximately 2 or more, preferably at least approximately 3 or more, further preferably at least approximately 5 or more, when cultured in the presence of hPL in a medium free of other animal-derived components, as compared with when cultured in a medium containing FBS.

[0171] The stem cell population of the present invention can reduce the amount of IL-6 production when cultured in the presence of hPL in a medium free of other animal-derived components, as compared with when cultured in a medium containing FBS.

[0172] The stem cell population of the present invention has high ability to produce SCF, ANGPTL4, and BIGH3 and a low production ratio of IL-6, and as such, is capable of safely exerting a tissue regenerating effect with low risk of inflammatory response.

1.4 Ability to Differentiate

[0173] The stem cell population of the present invention is multipotent and, as shown in Examples mentioned later, has the ability to differentiate into at least adipocytes, osteoblasts, chondrocytes, and neurons. The differentiation into the cells of interest can be induced as disclosed in Examples of the present specification or in accordance with a method known in the art (e.g., Marion et al., *Methods Enzymol.* (2006); 420:339-361; and Wang et al., *Molecular Medicine Reports* (2016); 14:5551-5555) by culturing the stem cell population of the present invention in the presence of a differentiation inducing factor appropriate for the cells of interest.

1.5 Physiological Activity

[0174] The stem cell population of the present invention produces a high level of a cytokine useful for tissue regeneration and has, in vivo or ex vivo, physiological activity (function) such as a neural progenitor cell proliferative effect, a neurite elongating effect, a vascular endothelial cell proliferative effect, a vascular endothelial cell recruiting effect, a vascular-like structure constructing effect, and an immunosuppressive effect.

2. Method for Producing Stem Cell Population of Present Invention

[0175] The stem cell population of the present invention can be produced by enzymatically isolating cells from the human dental pulp, culturing the cells in a medium free of FBS (fetal bovine serum) in the presence of a human platelet lysate (hPL), and obtaining a colony-forming cell population as the stem cell population.

[0176] The “dental pulp” used in the present invention may be any dental pulp of a deciduous tooth or a permanent tooth, and dental pulp of an extracted tooth such as a deciduous tooth or a wisdom tooth is preferred because of its easy availability. For the extracted tooth, it is desired to use a tooth within 72 hours after tooth extraction, more preferably within 48 hours after tooth extraction. It is particularly desired to use a human deciduous tooth preferably within 48 hours after tooth extraction. Examples of the method for enzymatically isolating the cell population from the collected dental pulp include a method of separating cells by treatment with an enzyme containing collagenase, preferably collagenase and neutral protease (e.g., dispase or thermolysin), and centrifuging the cells for isolation. For isolating the cell population from the collected dental pulp, it is preferred to perform treatment with a reagent containing no component derived from mammals and bacteria.

[0177] The isolated cells are cultured in the presence of a human platelet lysate (hPL) in a medium free of other animal-derived serum components, particularly, FBS. The “human platelet lysate (hPL)” is obtained by lysing platelet extracted from human blood by a freeze/thaw cycle, and abundantly contains various growth factors and cytokines necessary for cell culture. The amount of

the human platelet lysate (hPL) added to the medium or the content thereof in the medium is not particularly limited and is approximately 5 to 20%, preferably 5 to 15%, more preferably 7 to 15%, most preferably approximately 10% for primary culture. The same holds true for subculture. [0178] The term “approximately” as used herein refers to a value that varies within plus and minus 15%, 10%, 8%, 6%, 5%, 4%, 3%, 2%, or 1% based on a reference value. The term “approximately” preferably refers to the range of plus and minus 10%, 5%, or 1% based on a reference value.

[0179] The “medium” used in the present invention is not particularly limited as long as the medium is free of other animal-derived unpurified components, particularly, FBS. A serum-free medium is preferably used. The “serum-free medium” as used herein is a medium free of unadjusted or unpurified serum. A medium contaminated with a purified blood-derived component or animal tissue-derived factor (growth factor) is also included in the serum-free medium as long as it does not impair the object of the present invention. Examples of the serum-free medium can include basal media supplemented with no serum: for example, «MEM medium, DMEM medium, BME medium, Eagle MEM medium, BGJb medium, CMRL 1066 medium, Glasgow MEM (GMEM) medium, Improved MEM Zinc Option medium, IMDM medium, Medium 199 medium, F-12 medium, DMEM/F12 medium, IMDM/F12 medium, Ham's medium, RPMI 1640 medium, Fischer's medium, and mixed media thereof; and commercially available serum-free media for mammalian cells: for example, MesenCult-ACF (STEMCELL Technologies Inc.), STEMPRO MSC SFM (Thermo Fischer Scientific Inc.), and UltraCULTURE Serum-free (Lonza).

[0180] The “medium” for clinical application preferably contains no mammal-derived component and is particularly preferably animal-free.

[0181] The method for producing the stem cell population of the present invention is preferably a method using a human deciduous tooth within 48 hours after tooth extraction and comprising the steps of: i) enzymatically isolating a cell population from collected dental pulp using a reagent containing no component derived from mammals or bacteria; ii) primary-culturing the isolated cell population in the presence of a human platelet lysate (hPL) in a serum-free medium containing no other animal-derived unpurified components (particularly, FBS) to form colonies; and iii) culturing the obtained colonies in the presence of hPL in a serum-free medium containing no other animal-derived unpurified components (particularly, FBS).

[0182] The “medium” may be appropriately supplemented with various nutrient sources necessary for the maintenance and proliferation of the cells or each component necessary for the induction of differentiation without impairing the object of the present invention. Examples of the nutrient source can include carbon sources such as glycerol, glucose, fructose, sucrose, lactose, honey, starch, and dextrin, hydrocarbons such as fatty acids, fat or oil, lecithin, and alcohols, nitrogen sources such as ammonium sulfate, ammonium nitrate, ammonium chloride, urea, and sodium nitrate, inorganic salts such as common salt, potassium salt, phosphate, magnesium salt, calcium salt, iron salt, and manganese salt, monopotassium phosphate, dipotassium phosphate, magnesium sulfate, sodium chloride, ferrous sulfate, sodium molybdate, sodium tungstate, manganese sulfate, various vitamins, and amino acids.

[0183] The pH of the medium obtained by blending the components described above is in the range of 6.0 to 9.0, preferably 6.5 to 8.5, more preferably 7.0 to 8.0.

[0184] The cells isolated from the dental pulp, and the stem cell population of the present invention are adhesion-cultured. A container is not particularly limited as long as the container can be used for cell culture. A flask, a flask for tissue culture, a dish, a petri dish, a dish for tissue culture, a multidish, a microplate, a microwell plate, a multiplate, a multiwell plate, a microslide, a chamber slide, a petri dish, a tube, a tray, a culture bag, and a roller bottle can be used.

[0185] The inoculation density of the cells is not particularly limited and is preferably not too high. For example, cells extracted from the dental pulp of one deciduous tooth is inoculated at an area of 25 to 225 cm², preferably 75 to 150 cm². The culture is carried out at 36° C. to 38° C.,

preferably 36.5° C. to 37.5° C., under conditions of 1% to 25% O₂ and 1% to 15% CO₂ with the medium replaced.

[0186] The primary culture is preferably performed until colonies are formed and grow and detachment is confirmed. After colony detachment is confirmed, the cells are recovered using a filter or the like. The recovered cells are obtained as a highly homogeneous stem cell population without any special separation approach.

[0187] The stem cell population may be further subcultured (expansion-cultured), if necessary. The subculture is carried out using the same medium as that for primary culture. The passage is performed when 60% confluency is attained during adhesion culture. Typically, for the primarily cultured cells, the culture is performed for at least 7 days or longer, preferably 9 days or longer, more preferably 9 to 15 days, in order to confirm colony formation, growth, and detachment, whereas the cells thus passaged can be cultured for at least 1 day or longer, preferably 2 days or longer, more preferably 2 to 3 days. The inoculation density of the cells thus passaged is not particularly limited and is preferably not too high. The cells are inoculated at a density of, for example, 1000 to 2000 cells/cm², preferably 1300 to 1500 cells/cm².

[0188] The recovered stem cell population or a culture supernatant thereof is desirably subjected to an endotoxin test, a mycoplasma test, a sterilization test, or the like again for safety.

[0189] The produced stem cell population may be cryopreserved (e.g., preserved in a deep freezer of 152° C.), if necessary, until use. The cryopreservation is performed, for example, by adding an appropriate cryoprotectant to the medium used in the cell culture. Dextran, DMSO, a commercially available cryopreservation solution, or the like can be used as the cryoprotectant.

3. Culture Supernatant of Stem Cell Population of Present Invention

[0190] The culture supernatant of the stem cell population of the present invention abundantly contains a cytokine useful for tissue regeneration and can be used in itself as a pharmaceutical composition for tissue regeneration, etc.

[0191] Specifically, the culture supernatant of the stem cell population of the present invention has at least one of the following characteristics 1) to 3): [0192] 1) comprising SCF at a weight ratio of 1/20 or more, preferably 1/10 or more, more preferably 1/10 to 1/2, further preferably 1/10 to 1/3, relative to IL-6 (however, without exceeding the amount of production of IL-6), [0193] 2) comprising ANGPTL4 at a weight ratio of 3 or more, preferably 5 or more, more preferably 5 to 20, further preferably 5 to 15, relative to IL-6, [0194] 3) comprising BIGH3 at a weight ratio of 400 or more, preferably 450 or more, more preferably 450 to 1500, further preferably 450 to 1000, relative to IL-6.

[0195] The culture supernatant of the stem cell population of the present invention particularly preferably has at least one of the following characteristics 1') to 3'): [0196] 1') comprising SCF at a weight ratio of 1/20 or more, preferably 1/10 or more, more preferably 1/5 or more, further preferably 1/3 or more, relative to IL-6 (however, without exceeding the amount of production of IL-6), [0197] 2') comprising ANGPTL4 at a weight ratio of 3 or more, preferably 5 or more, more preferably 10 or more, further preferably 20 or more, relative to IL-6, and [0198] 3') comprising BIGH3 at a weight ratio of 400 or more, preferably 450 or more, more preferably 450 to 1500, further preferably 450 to 1000, relative to IL-6.

[0199] The culture supernatant of the present invention can be cryopreserved until clinical use and can be used in the treatment of various diseases in need of tissue regeneration, in the same way as a pharmaceutical composition mentioned later.

4. Pharmaceutical Composition Comprising Stem Cell Population of Present Invention

[0200] The stem cell population of the present invention abundantly produces a cytokine useful for tissue regeneration, such as SCF, has a low production ratio of an inflammatory cytokine, and is CD117-negative. Therefore, the stem cell population of the present invention has an excellent tissue regenerating effect and can be used as a pharmaceutical composition. The stem cell population of the present invention is provided as a cell medicament, if necessary, together with a

culture medium or a culture supernatant.

[0201] The pharmaceutical composition of the present invention has at least one of the following characteristics 1) to 3): [0202] 1) comprising SCF at a weight ratio of 1/20 or more, preferably 1/10 or more, more preferably 1/10 to 1/2, further preferably 1/10 to 1/3, relative to IL-6 (however, without exceeding the amount of production of IL-6), [0203] 2) comprising ANGPTL4 at a weight ratio of 3 or more, preferably 5 or more, more preferably 5 to 20, further preferably 5 to 15, relative to IL-6, [0204] 3) comprising BIGH3 at a weight ratio of 400 or more, preferably 450 or more, more preferably 450 to 1500, further preferably 450 to 1000, relative to IL-6.

[0205] The pharmaceutical composition of the present invention preferably has at least one of the following characteristics 1') to 3'): [0206] 1) comprising SCF at a weight ratio of 1/20 or more, preferably 1/10 or more, more preferably 1/5 or more, further preferably 1/3 or more, relative to IL-6, [0207] 2) comprising ANGPTL4 at a weight ratio of 3 or more, preferably 5 or more, more preferably 10 or more, further preferably 20 or more, relative to IL-6, [0208] 3) comprising BIGH3 at a weight ratio of 400 or more, preferably 450 or more, more preferably 450 to 1500, further preferably 450 to 1000, relative to IL-6.

[0209] The number of stem cells contained in the pharmaceutical composition of the present invention is appropriately determined depending on a subject or a target disease. A minimum effective amount is practical in consideration of the timing of administration to a subject and time required for culture. In general, the number of cells contained in the pharmaceutical composition is 1×10^5 or more cells or 1×10^6 or more cells, preferably 1×10^7 or more cells, more preferably 1×10^8 or more cells, further preferably 1×10^9 or more cells. The number of doses is not limited to one, and the pharmaceutical composition may be administered two or more times.

[0210] The pharmaceutical composition of the present invention is preferably a parenteral preparation, more preferably a parenteral systemic preparation, particularly, an intravenous preparation. Examples of the dosage form suitable for parenteral administration include injections such as solution-type injections, suspension-type injections, emulsion-type injections, and injections which are prepared upon use, and grafts. The preparation for parenteral administration is in the form of an aqueous or nonaqueous isotonic sterile solution or suspension and is formulated into an appropriate unit dosage form, for example, by appropriately combining pharmaceutically acceptable carriers or vehicles, specifically, sterile water or physiological saline, a medium (particularly, a medium for use in the culture of mammalian cells, such as RPMI), a physiological buffer solution such as PBS, a plant oil, an emulsifier, a suspending agent, a surfactant, a stabilizer, an excipient, a vehicle, an antiseptic, and a binder.

[0211] The disease to which the pharmaceutical composition of the present invention is applicable is not particularly limited as long as the disease is in need of tissue regeneration. Examples thereof can include spinal cord injury (including traumatic injury, injury caused by surgical operation, and chronic-phase spinal cord injury), ischemic disease (cerebral infarction, limb ischemia including lower limb ischemia, perinatal hypoxic ischemic encephalopathy, ischemic heart disease including myocardial infarction, etc.), inflammatory disease (sepsis, hepatitis, pancreatitis, nephritis, pneumonitis, etc.), autoimmune disease (rheumatism, SLE, type I diabetes mellitus, etc.), intestinal disease (irritable bowel syndrome, ulcerative colitis, Crohn's disease, Hirschsprung's disease and related syndromes, etc.), and the treatment or regeneration of defects in bone.

5. Treatment Method Using Stem Cell Population of the Present Invention or Culture Supernatant Thereof

[0212] The present invention also provides a treatment method comprising administering the stem cell population of the present invention or the culture supernatant thereof to a subject in need. The subject to be treated is not particularly limited as long as the subject is in need of tissue regeneration. Examples thereof can include the above-described spinal cord injury, ischemic disease, inflammatory disease, autoimmune disease, and intestinal disease, and these diseases may

be in an acute phase, a subacute phase, or a chronic phase. The subject to be treated by the treatment method of the present invention may be, for example, acute-phase or subacute-phase spinal cord injury, chronic-phase ischemic disease, chronic-phase inflammatory disease, chronic-phase autoimmune disease, chronic-phase intestinal disease, acute-phase or subacute-phase bone defect, or chronic-phase bone defect.

6. Dental Pulp Stem Cell Bank

[0213] A dental pulp stem cell bank can be prepared by preparing dental pulp-derived stem cell populations from a plurality of donors in accordance with the method of the present invention, and cryopreserving the stem cell populations. A method for cryopreserving the cells can be carried out by a method known in the art, as mentioned above.

7. Agent for Preventing or Treating Cerebral Palsy

[0214] In one aspect, the present invention relates to an agent for preventing or treating cerebral palsy, comprising dental pulp stem cells. In another aspect, the present invention relates to an agent for promoting neurogenesis in the brain of a subject with cerebral palsy symptoms, comprising dental pulp stem cells (in the present specification, these agents are also collectively referred to as the “prevention or treatment agent of the present invention”). Hereinafter, these agents will be described.

[0215] The dental pulp stem cells are not particularly limited as long as the stem cells are obtained from dental pulp (derived from dental pulp). Examples of the dental pulp stem cells include deciduous dental pulp stem cells (dental pulp stem cells derived from deciduous teeth) and permanent dental pulp stem cells (dental pulp stem cells derived from permanent teeth). The dental pulp stem cells are stem cells derived from low invasively and easily collectable neural crest and are a cell population coexpressing various undifferentiated nervous system markers and mesenchymal stem cell markers (Miura M. et al.; Proc Natl Acad Sci USA. 2003 May 13; 100 (10): 5807-123). The dental pulp stem cells are preferably deciduous dental pulp stem cells from the viewpoint of a prophylactic or therapeutic effect on cerebral palsy, etc., particularly preferably deciduous dental pulp stem cells collected from a deciduous tooth within 48 hours after fallout or tooth extraction.

[0216] Examples of the organism from which the dental pulp stem cells are derived include, but are not particularly limited to, various mammals such as humans, monkeys, mice, rats, dogs, cats, rabbits, pigs, horses, bovines, sheep, goats, and deer. The dental pulp stem cells are preferably human dental pulp stem cells, further preferably human deciduous dental pulp stem cells, particularly preferably deciduous dental pulp stem cells obtained from a deciduous tooth within 48 hours after fallout or tooth extraction.

[0217] In relation to a subject (patient, etc.) to which the prevention or treatment agent of the present invention is applied, the dental pulp stem cells are preferably dental pulp stem cells of the same organism species (i.e., derived from a human if the subject is a human) in order to suppress or circumvent rejection. The dental pulp stem cells can be autologous dental pulp stem cells or allogeneic dental pulp stem cells. The “prevention” includes a state in which the onset of symptoms or decline in physical function is suppressed or slowed down. The “treatment” includes the alleviation of symptoms or improvement in declined physical function. Examples of such symptoms or functions include motor functions, posture, memory, and learning for cerebral palsy.

[0218] The dental pulp stem cells can be selected as adherent cells among dental pulp cells. Adherent cells among dental pulp cells collected from a deciduous tooth that has fallen out or a permanent tooth, or passaged cells thereof can be used as the “dental pulp stem cells”. The dental pulp stem cells can be prepared by, for example, the following method described in Japanese Patent Laid-Open No. 2011-219432.

(1) Collection of Dental Pulp

[0219] A deciduous tooth that has fallen out spontaneously (or an extracted deciduous tooth or permanent tooth) is disinfected with a chlorhexidine or povidone-iodine solution. Then, the crown

of the tooth is divided, and a dental pulp tissue is recovered using a dental reamer.

(2) Enzymatic Treatment

[0220] The collected dental pulp tissue was suspended in a basal medium (e.g., MEM α) and treated with 0.02 to 2 mg/ml tissue-degrading enzyme (e.g., collagenase or dispase). The dental pulp cells thus enzymatically treated are recovered by centrifugal operation.

(3) Cell Culture (Selection of Adherent Cell)

[0221] The cells are resuspended in a basal medium and inoculated to a dish for adhesion cell culture. The cells are cultured in an incubator adjusted to 5% CO₂ and 37° C. and then washed, if necessary, and adherent cells that have formed colonies are then treated with 0.05% trypsin-EDTA at 37° C. for 5 minutes. The dental pulp cells detached from the dish are washed, if necessary, then inoculated to a dish for adhesion cell culture, and expansion-cultured. The cells, when reaching subconfluency (state in which the cells occupy approximately 70% of the surface of the culture container) or confluency in macroscopic observation, are detached and recovered from the culture container and inoculated again to a culture container filled with a culture medium. Subculture may be repetitively performed. For example, the cells are allowed to proliferate into a necessary number of cells (e.g., approximately 1×10^7 cells/ml) by performing subculture 1 to 8 times. The detachment of the cells from the culture container can be carried out by a routine method such as trypsin treatment. The cells thus cultured may be recovered and preserved (preservation conditions involve, for example, -150° C.).

[0222] For example, a basal medium or a basal medium supplemented with serum or the like can be used as the culture solution. DMEM as well as Iscove's modified Dulbecco's medium (IMDM) (Gibco, etc.), Ham's F12 medium (HamF12) (Sigma-Aldrich Co., LLC, Gibco, etc.), RPMI1640 medium, or the like can be used as the basal medium. Two or more basal media may be used in combination. One example of the mixed medium can include a medium of IMDM and HamF12 mixed in equal amounts (e.g., commercially available under a trade name of IMDM/HamF12 (Gibco)). Examples of the component that may be added to the medium can include serum (fetal bovine serum, human serum, sheep serum, etc.), serum replacements (knockout serum replacement (KSR), etc.), bovine serum albumin (BSA), antibiotics, various vitamins, and various minerals.

(4) Recovery of Cell

[0223] The cells can be recovered by detaching the cells from the culture container by trypsin treatment or the like, followed by centrifugation. The prevention or treatment agent of the present invention can be prepared using the dental pulp stem cells thus recovered.

[0224] In an embodiment, the dental pulp stem cells are a non-transgenic stem cell population derived from human deciduous dental pulp, at least 90% or more of which is CD117-negative, CD73-positive, CD90-positive, and CD105-positive (hereinafter, also referred to as the "stem cell population of the present invention"). Structural or functional characteristics of the stem cell population of the present invention is described in items 1.2 to 1.5.

[0225] The dental pulp stem cells have a prophylactic or therapeutic effect on cerebral palsy and a promoting effect on neurogenesis in the brain. Hence, the dental pulp stem cells can be used as an active ingredient in an agent for preventing or treating cerebral palsy or an agent for promoting neurogenesis. The dental pulp stem cells used in the present invention may be used as, for example, a medicament. The dental pulp stem cells may be used together with an additional active ingredient, and, preferably, the dental pulp stem cells are used as a single active ingredient.

[0226] Preferably, the dental pulp stem cells used in the present invention are cells that have not undergone induction of differentiation, particularly preferably cells that have not undergone induction of differentiation into neurons. The dental pulp stem cells can be genetically engineered by various known methods. The dental pulp stem cells used in the present invention are preferably non-transgenic cells.

[0227] Cerebral palsy is lifelong and variable abnormality of movement, posture, or memory learning in children after birth based on a nonprogressive lesion including neuronal death in the

children's brain caused from conception to the neonatal stage (within 4 weeks after birth). However, progressive disease, transient dyskinesia, or motor retardation that seems to return normal in the future is excluded therefrom. Although symptoms or severity of the disorder differs among individuals depending on a lesion site in the brain, there exists classification such as limb rigidity (spastic type), excessive limb movement (athetotic type), and decreased muscle tonus (dyskinetic type). Mental retardation, epilepsy, indirect contracture, and scoliosis are known as complications.

[0228] Cerebral palsy is caused by brain malformation or prenatal infection before birth, hypoglycemia, intracranial bleeding, premature birth, or neonatal asphyxia (HIE) at the perinatal stage, and infection of encephalitis, meningitis, or the like for newborns. Among others, cerebral palsy is often caused by perinatal disorder (perinatal cerebral damage).

[0229] The type of the cerebral palsy is not particularly limited. Cerebral palsy ascribable to various causes is to be prevented or treated. The dental pulp stem cells are preferably effective for preventing the manifestation of cerebral palsy symptoms caused by perinatal cerebral damage, particularly, symptoms of cerebral palsy caused by hypoxia and/or ischemia, and for treating these symptoms. The treatment may be provided, for example, when a perinatal infant has undergone neonatal asphyxia and/or been diagnosed with periventricular leukomalacia, or may be provided at the chronic stage (6 months or later after birth), and is particularly preferably provided to chronic stage cerebral palsy. The prevention or treatment agent of the present invention may be prophylactically used for, for example, a subject suspected of having cerebral palsy (e.g., a subject who has undergone neonatal asphyxia and/or been diagnosed with periventricular leukomalacia at the perinatal stage) or may be used for suppressing or improving the progression of symptoms in a subject with cerebral palsy symptoms. The prevention or treatment agent of the present invention can be preferably used for improving a motor function and/or a memory learning function in a subject with cerebral palsy symptoms and/or for promoting neurogenesis in the brain of the subject. More preferably, the prevention or treatment agent of the present invention can be used for improving a motor function and a memory learning function in a subject with cerebral palsy symptoms. Further preferably, the prevention or treatment agent of the present invention can be used for improving a motor function and a learning function in a subject with cerebral palsy symptoms and for promoting neurogenesis in the brain of the subject.

[0230] Neurogenesis is a phenomenon in which neurons are newly produced from neural stem cells. Neurogenesis is known to occur in particular regions such as the hippocampal dentate gyrus or the subventricular zone not only at the stage of mammalian development but in adults (Gage, J. of Neuroscience, Feb. 1, 2002, 22 (3): 612-613). It is known that when cerebral damage such as perinatal damage occurs, neural stem cells of the subventricular zone or the hippocampal dentate gyrus proliferate and lead to a compensation mechanism in which some of the cells migrate to the brain lesion, whereas the proliferation or migration of such neural stem cells is reduced after a lapse of a given time (approximately 1 month on average in the observation of animal models) after damage (Visco et al., Experimental Neurology 340 (2021) 113643). Hence, the cerebral damage of cerebral palsy cannot be repaired by only the compensation mechanism possessed by organisms as mentioned above (Visco et al., Experimental Neurology 340 (2021) 113643, Chen et al., Frontiers in Pediatrics (2022) 10:986452).

[0231] In the present application, the phrase "promote neurogenesis" by the prevention or treatment of the present invention means that the proliferation of neural stem cells and/or increase in the number of neurons in a brain tissue (particularly, the subventricular zone or the hippocampal dentate gyrus as mentioned above) in a subject with cerebral palsy is markedly enhanced as compared with the case where the prevention or treatment agent of the present invention is not administered.

[0232] Neurogenesis in the brain can be detected by combining markers such as bromodeoxyuridine (BrdU), doublecortin (DCX), and Ki-67. NeuN can detect mature neurons in

brain tissues.

[0233] The prevention or treatment agent of the present invention may be administered to, for example, a subject suspected of having cerebral palsy without manifesting symptoms of cerebral palsy, may be administered to a subject with symptoms of cerebral palsy, or may be administered to a subject diagnosed with cerebral palsy. Examples of the subject suspected of having cerebral palsy include subjects who have undergone a hypoxic state at the perinatal stage, subjects who have undergone ischemia, subjects who have undergone neonatal asphyxia, and subjects who have been diagnosed with periventricular leukomalacia. In the present application, the “subject with symptoms of cerebral palsy” is a subject who manifests one or more symptoms of cerebral palsy and may have already been diagnosed with cerebral palsy or may be before diagnosis.

[0234] The prevention or treatment agent of the present invention is not particularly limited as long as the prevention or treatment agent contains the active ingredient. The prevention or treatment agent of the present invention may optionally further contain an additional component. The additional component is not particularly limited as long as the component is pharmaceutically acceptable. The additional component includes a component having a pharmacological effect as well as an additive. Examples of the additional component include carriers or vehicles, excipients, disintegrants, buffers, emulsifiers, suspending agents, soothing agents, stabilizers, preservatives, antiseptics, physiological saline, albumin, media, antibiotics, pH adjusters, growth factors, and hormones. Examples of the carriers or vehicles include physiological saline, Ringer's solutions, and appropriate buffer solutions (e.g., phosphate buffer solutions).

[0235] Examples of the subject to which the prevention or treatment agent of the present invention is applied include, but are not particularly limited to, mammals, for example, humans, monkeys, mice, rats, dogs, cats, rabbits, pigs, horses, bovines, sheep, goats, and deer.

[0236] The subject to which the prevention or treatment agent of the present invention is applied is preferably a human. In the case of a human, the prevention or treatment agent of the present invention can be used so that the agent is administered from infancy onward (1 year old or later, for example, 2 years old or later, 3 years old or later, 4 years old or later, 5 years old or later, or 6 years old or later). The subject to which the prevention or treatment agent of the present invention is applied can be a subject with symptoms of cerebral palsy, for example, a subject diagnosed as having symptoms of cerebral palsy. The prevention or treatment agent of the present invention can exert a therapeutic effect or the like on cerebral palsy even if administered to the subject.

[0237] Examples of the administration route of the prevention or treatment agent of the present invention include intravenous injection (including intravenous drip), intra-arterial injection, intraportal injection, intradermal injection, subcutaneous injection, intramuscular injection, intraperitoneal injection, and transnasal administration. The prevention or treatment agent of the present invention may be locally administered instead of systemic administration. Examples of the local administration can include direct injection to brain tissues. In the case of administering the prevention or treatment agent of the present invention as an injection, the dental pulp stem cells can be suspended, for administration, in a cell culture medium, physiological saline, a composite electrolyte liquid, or the like. Such a suspension may contain serum, albumin, dextran, dimethyl sulfoxide, a cryopreservative solution, or the like.

[0238] The content of the dental pulp stem cells in the prevention or treatment agent of the present invention and the dosing schedule thereof can be designed in consideration of the sex, age, body weight, pathological condition, etc. of a subject (patient). The content of the dental pulp stem cells can be, for example, 1×10^5 to 1×10^{11} cells/kg body weight per dose. Single-dose administration as well as continuous or periodic multiple-dose administration may be performed. An appropriate number of doses is 2 to 20 doses (preferably 2 to 10 doses or 2 to 5 doses). The dosing interval can be selected from 1 day to 4 weeks and is preferably 7 days to 24 days or 10 days to 18 days. The multiple-dose administration can more markedly improve a motor function and/or a memory learning function in a subject with cerebral palsy symptoms and/or markedly

promote neurogenesis in the brain thereof.

EXAMPLES

[0239] Hereinafter, the present invention will be described more specifically with reference to Examples. However, the present invention is not limited by these Examples.

Example 1: Preparation of Dental Pulp Stem Cell

[0240] In a sterile environment, dental pulp was scraped out of an extracted tooth (human deciduous tooth) within 48 hours after tooth extraction, chopped, and then treated with Liberase (MNP-S GMP Grade, Roche) added at a final concentration of 0.05 mg/ml with stirring at 37° C. for 15 minutes. A reaction supernatant containing cells was recovered, and the remaining dental pulp tissue was further treated with Liberase added at 0.05 mg/ml with stirring at 37° C. for 15 minutes. Cells were recovered from the treated dental pulp tissue through a 70 mm strainer, combined with the supernatant recovered first, and centrifuged. The recovered cells were inoculated to a culture container (CellBind T75 flask) using MEMX (Gibco) containing 10% hPL (AventaCell BioMedical Co., Ltd.), and isolated and cultured for 9 days or longer until colonies were formed, grew, and were detached. Medium replacement was performed 2 days and 9 days after inoculation. After colony detachment was confirmed, cells were recovered using TrypLE Select (Gibco) and expansion-cultured using the same medium as above. Hereinafter, human dental pulp stem cells prepared by this method are also referred to as the present human dental pulp stem cells.

Example 2: Surface Marker Analysis (1)

[0241] Human dental pulp stem cells prepared from 8 donors in accordance with Example 1 were reacted at 1×10^5 cells per donor with phycoerythrin (PE)-labeled antibodies against various surface markers (CD117 (#313204), CD31 (#303105), CD34 (#343606), CD45 (#368510), CD90 (#328110), CD105 (#323206), CD73 (#344004), CD325 (#350806), CD49d (#304303), CD51 (#327909), CD29 (#303003), CD44 (#338807), CD166 (#343904), and CD106 (#305805), all manufactured by BioLegend, Inc.), and data was obtained by use of flow cytometry. Analysis was conducted by gating such that an isotype control antibody-positive ratio would be around 1%, and by calculating positive ratios for the various surface markers.

TABLE-US-00001 TABLE 1 Positive ratio for surface marker (%)													
Donor 1	Donor 2	Donor 3	Donor 4	Donor 5	Donor 6	Donor 7	Donor 8	CD117	CD31	CD34	CD45	CD90	CD105
1.99	1.82	1.36	1.36	0.58	1.10	0.91	1.20	4.16	6.12	1.32	5.37	2.20	3.13
2.06	15.49	1.54	1.52	0.17	0.72	6.86	19.96	1.31	3.29	0.97	0.92	0.22	1.36
0.46	0.90	0.56	0.51	100.00	100.00	99.99	100.00	100.00	100.00	99.99	100.00	100.00	99.99
100.00	100.00	100.00	100.00	100.00	100.00	99.99	100.00	99.99	100.00	99.99	100.00	100.00	99.99
2.31	1.68	2.04	3.28	4.44	1.74	1.40	4.10	96.10	99.60	99.22	99.98	99.99	99.60
99.91	99.99	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	100.00	100.00	100.00	100.00

[0242] As a result, all the donor cells had a positive ratio of less than 5% for CD117, CD325, and CD45 and were thus determined as being negative therefor. All the donor cells had a positive ratio of 96% or more for CD73, CD90, CD105, CD49d, and CD51 and were thus determined as being positive therefor. The cells of 7 out of the 8 donors had a positive ratio of less than 7% for CD31 and CD34 and were thus determined as being negative therefor. The cells of one donor exhibited a CD31-positive ratio of 15.49%, and the cells of another donor exhibited a CD34-positive ratio of 19.96% (FIG. 1A and FIG. 1B). All the donor cells had a positive ratio of 99% or more for CD29, CD44, and CD166. A CD106-positive ratio was less than 9% in the cells of 7 out of the 8 donors and 10.75% in the cells of the remaining one donor.

Example 3: Ability to Differentiate into Adipocyte

[0243] The present human dental pulp stem cells obtained in the same manner as in Example 1 were inoculated at 5.76×10^5 cells/2.5 mL/well to a 6-well plate using 10% hPL/MEM α , and cultured until reaching 100% confluency, followed by replacement with an adipose differentiation medium (MEM α , 2% FBS, 500 μ M IBMX, 50 μ M indomethacin, 5 μ g/ml insulin, and 1 μ M

dexamethasone). After differentiation was induced for 28 days, the cells were stained using a lipid assay kit (Cosmo Bio Co., Ltd., #AK09F), and a fat droplet image was observed under a microscope.

[0244] As a result, fat droplets stained red were found in the cells (indicated by the arrow in the drawing), demonstrating that the present human dental pulp stem cells have the ability to differentiate into adipocytes (FIG. 2).

Example 4: Ability to Differentiate into Osteoblast

[0245] The present human dental pulp stem cells obtained in the same manner as in Example 1 were inoculated at 5.76×10^5 cells/2.5 mL/well to a collagen-coated 6-well plate using 10% hPL/MEM α , and cultured until reaching 100% confluency, followed by replacement with a bone differentiation medium (MEMA, 5% FBS, 50 μ g of L-ascorbic acid 2-phosphate, 10 nM dexamethasone, and 10 mM β -glycerophosphate). After differentiation was induced for 28 days, the cells were stained using a calcification staining kit (Cosmo Bio Co., Ltd., #AK21), and a calcification image was observed under a microscope.

[0246] As a result, a calcification image stained red (indicated by the arrow in the drawing) was found, demonstrating that the present human dental pulp stem cells have the ability to differentiate into osteoblasts (FIG. 3).

Example 5: Ability to Differentiate into Chondrocyte

[0247] The present human dental pulp stem cells (donor 1) obtained in Example 1 were inoculated at 2×10^5 cells/200 L/well to a low-adsorption 96 U-bottom plate (Sumitomo Bakelite Co., Ltd., SUMS9096U) using 10% hPL/MEM α . Spheroid formation was confirmed on the next day, followed by replacement with a cartilage differentiation medium (PromoCell, #C-28012). After differentiation was induced for 21 days, the cells were fixed in 4% paraformaldehyde, embedded in paraffin blocks, and then sliced. The slices were stained using an alcian blue staining solution, and a cartilage tissue image was observed under a microscope.

[0248] As a result, a cartilage tissue image stained blue (indicated by the arrow in the drawing) was found, demonstrating that the present human dental pulp stem cells have the ability to differentiate into chondrocytes (FIG. 4).

Example 6: Ability to Differentiate into Neuron

[0249] The present human dental pulp stem cells obtained in the same manner as in Example 1 were inoculated at 2.8×10^3 cells/400 μ L/well to poly-D-lysine/laminin cell ware 8-well culture slide (Corning Inc.) using 10% hPL/MEM α , followed by replacement with N.sub.2 medium (Neurobasal-A+N2 supplement serum free, 10 ng/ml human EGF, and 10 ng/ml human FGF-basic) 24 hours later. After differentiation was induced for 21 days, the cells were fixed in 4% paraformaldehyde, stained using anti- β III-tubulin (clone: TU-20) (Millipore, #MAB1637), anti-nestin (clone: 10C2) (Novus Biologicals, LLC, #NB300-266), and anti-neurofilament M (clone: NN18) (Sigma, #5264) antibodies as primary antibodies against neural differentiation markers and Alexa Fluor® 488 anti-mouse IgG (H+L) (Jackson ImmunoResearch Laboratories, Inc.) as a secondary antibody, and observed under a fluorescence microscope.

[0250] As a result, all the neural differentiation markers were found on the basis of cells immunostained with the antibodies thereagainst, demonstrating that the present human dental pulp stem cells have the ability to differentiate into neurons (FIG. 5).

Example 7: Ability to Produce Cytokine (1)

[0251] The present human dental pulp stem cells were subcultured in 10% hPL/MEM α or 10% FBS/MEM α for 7 days and then inoculated at 5.76×10^5 cells/2.5 ml/well to a 6-well plate using the same medium as above, followed by replacement with hPL-free, FBS-free, and phenol red-free MEM α at 2.5 ml/well on the next day. A culture supernatant was recovered 48 hours later, dispensed, and stored at -80° C. Frozen samples were thawed. SCF, ANGPTL4, BIGH3, and IL-6 concentrations in the samples were measured using Multiplex HGF panel (BioLegend, Inc., #740180), Human ANGPTL4 assay kit (IBL, #27749), Human beta IG-H3 ELISA (Abcam plc,

#ab220651), and Multiplex Neuroinflammatory panel (BioLegend, Inc., #740796), respectively, and their respective concentration ratios to IL-6 were calculated (Tables 2 and 3). From the amount of the culture supernatant and the number of inoculated cells, the amounts of various factors contained in the cell culture medium were calculated as the amounts of production per 1×10^6 cells (Tables 4 and 5).

[0252] As a result, the culture supernatant of the dental pulp stem cells subcultured in 10% hPL/MEM had higher concentration ratios to IL-6 and amounts of production per 1×10^6 cells for all of SCF, ANGPTL4, and BIGH3 than those of the culture supernatant of the dental pulp stem cells subcultured in 10% FBS/MEMA.

TABLE-US-00002 TABLE 2 Concentrations and ratios of SCF, ANGPTL4, BIGH3, and IL-6 produced by human dental pulp stem cell cultured in 10% hPL/MEM α

Concentration in culture supernatant (ng/mL)	Ratio to IL-6	Donor 1	Donor 2	Donor 1	Donor 2	n = 3	Mean	n = 3	Mean
Mean	Mean	SCF	0.0406	0.0395	0.0767	0.0606	0.29	0.21	0.0396
0.0598	0.0383	0.0452	ANGPTL4	1.36	1.61	2.61	3.41	12	12
1.68	3.92	1.79	3.69	BIGH3	71.5	128	123	140	944
487	166	130	147	167	IL-6	0.167	0.136	0.266	0.288
0.122	0.261	0.119	0.337						

TABLE-US-00003 TABLE 3 Concentrations and ratios of SCF, ANGPTL4, BIGH3, and IL-6 produced by human dental pulp stem cell cultured in 10% FBS/MEM α

Concentration in culture supernatant (ng/mL)	Ratio to IL-6	Donor 1	Donor 2	Donor 1	Donor 2	n = 3	Mean	n = 3	Mean
Mean	Mean	SCF	0.0125	0.0117	0.0135	0.0145	0.05	0.04	0.0112
0.0153	0.0114	0.0149	ANGPTL4	0.417	0.574	0.143	0.141	2.3	0.36
0.592	0.143	0.712	0.138	BIGH3	71.2	77.2	54.2	53.5	309
135	58.3	58.3	101.9	48.0	IL-6	0.350	0.249	0.375	0.397
0.196	0.486	0.202	0.331						

TABLE-US-00004 TABLE 4 Amounts of SCF, ANGPTL4, BIGH3, and IL-6 produced by human dental pulp-derived cell cultured in 10% hPL/MEM α

Amount of secretion into culture supernatant (ng/ 10^6 cells)	Donor 1	Donor 2	n = 3	Mean	n = 3	Mean
SCF	0.176	0.171	0.333	0.263	0.172	
ANGPTL4	5.89	6.99	11.3	14.8	7.30	
17.0	7.78	16.0	BIGH3	311	557	
535	609	723	564	639	727	
IL-6	0.724	0.591	1.16	1.25	0.531	
1.13	0.516	1.46				

TABLE-US-00005 TABLE 5 Amounts of SCF, ANGPTL4, BIGH3, and IL-6 produced by human dental pulp-derived cell cultured in 10% FBS/MEM α

Amount of secretion into culture supernatant (ng/ 10^6 cells)	Donor 1	Donor 2	n = 3	Mean	n = 3	Mean
SCF	0.0542	0.0508	0.0584	0.0631	0.0486	
ANGPTL4	1.81	2.49	0.619	0.613	2.57	
0.619	3.09	0.600	BIGH3	309	335	
235	232	253	253	442	208	
IL-6	1.52	1.08	1.63	1.72	0.852	
2.11	0.877	1.44				

Example 8: Neurite Elongating Effect

[0253] The present human dental pulp stem cells (donor 8) were inoculated at a density of 1000 cells/cm² using 10% hPL/MEM α and cultured for 6 days, followed by medium replacement with MEM α . After culture for another 7 days, a culture supernatant thereof was recovered. This culture supernatant was added to a poly-D lysin (PDL)-coated culture plate (Corning Inc.) so that the plate was coated with components contained in the culture supernatant. IMR-32 cells were inoculated to this plate and induced to differentiate into neurons by the addition of all trans retinal on Days 1 to 4 and Days 7 to 8. On Day 9, neurites and nuclei were stained with Tubulin Tracker™ Green (Thermo Fisher Scientific Inc.) and Hoechst 33342 (Dojindo Laboratories), respectively, and photographed, followed by neurite length measurement by image analysis using Autoneurite J.

[0254] As a result, the lengths of neurites elongating from the IMR-32 cells induced to differentiate into nerve were significantly larger in the plate coated with the culture supernatant than in the plate coated with only PDL. This suggested that neurite elongation was promoted by extracellular matrix or the like secreted by the present human dental pulp stem cells (FIG. 6). This effect is important for nerve regeneration.

Example 9: Neural Progenitor Cell Proliferative Effect

[0255] Human neural progenitor cells (ENStemA, Millipore) were suspended (5×10^4 cells/100 μ l/well in a 96-well plate) in ENstem expansion medium containing L-glutamine and FGF and inoculated to Corning BioCoat poly-L-ornithine/laminin-coated multiwell plate. On the next day,

the medium was replaced with L-glutamine-containing ENstem expansion medium supplemented with either the dental pulp stem cell culture supernatant recovered in Example 8 or MEMA (control) at a ratio of 20% or 40%, and the cells were further cultured for 3 days. CCK-8 was added to the neural progenitor cells, and the number of neural progenitor cells was evaluated by measuring absorbance at 450 nm 1 hour later.

[0256] As a result, absorbance was decreased in a manner dependent on the amount of MEMA added in the neural progenitor cells supplemented with MEMA, whereas absorbance was rather elevated in the neural progenitor cells supplemented with the culture supernatant. This suggested that neural progenitor cells proliferated in response to a nutritional factor or the like secreted by the present human dental pulp stem cells (FIG. 7). This effect is important for nerve regeneration.

Example 10: Vascular Endothelial Cell Proliferative Effect (1)

[0257] HUVEC (human umbilical vein endothelial cell line, Kurabo Industries Ltd.) was suspended in Humedia/2% FBS and then inoculated (2.5×10^3 cells/well) to a 96-well plate. On the next day, the dental pulp stem cell culture supernatant recovered in Example 8, which was mixed with MEMA at various ratios and further supplemented with 2% (final concentration) FBS, was added in equal amounts from above Humedia/2% FBS in the wells, and the cells were cultured for 3 days. CCK-8 was added to HUVEC and reacted at 37° C. for 4 hours. Then, the number of HUVEC cells was evaluated by measuring OD450.

[0258] As a result, absorbance was elevated in a manner dependent on the ratio of the culture supernatant. This suggested that vascular endothelial cells proliferated in response to a nutritional factor or the like secreted by the present human dental pulp stem cells (FIG. 8). This effect is important for angiogenesis.

Example 11: Vascular-Like Structure Constructing Effect (1)

[0259] HUVEC suspended in the dental pulp stem cell culture supernatant recovered in Example 8 or MEMA was inoculated at 1×10^5 cells/well to Extracellular matrix gel (Angiogenesis Assay Kit, Abcam plc) prepared in a 96-well plate, and cultured for 20 hours. The wells were photographed in the bright field, and a branching interval was measured by image analysis using Angiogenesis Analyzer for ImageJ to study a tube forming (vascular-like structure constructing) effect.

[0260] As a result, a branching interval was markedly increased by the addition of the culture supernatant (FIG. 9). This suggested that tube formation was promoted by a nutritional factor or the like secreted by the present human dental pulp stem cells. This effect is important for angiogenesis.

Example 12: Vascular Endothelial Cell Recruiting Effect

[0261] HUVEC was inoculated (1×10^5 cells/well) to the upper layer of Transwell. The lower layer was filled with the dental pulp stem cell culture supernatant recovered in Example 8 or MEMA, and the cells were cultured for 24 hours. HUVEC moved to the back side (lower layer side) of the upper layer was detached with trypsin. Then, the number of cells thereof was measured as luminescence intensity with CellTiterGlo (Promega Corp.) to evaluate an effect in which HUVEC were recruited to the lower layer.

[0262] As a result, luminescence intensity was markedly increased by filling the lower layer with the culture supernatant (FIG. 10). This suggested that vascular endothelial cells were strongly recruited by a chemokine or the like secreted by the present human dental pulp stem cells. This effect is important for angiogenesis.

Example 13: Immunosuppressive Effect

[0263] The present human dental pulp stem cells (1×10^5 cells/well in a 96-well plate) were mixed with CFSE (dye for live cell staining, Thermo Fisher Scientific Inc.)-labeled human PBMC (peripheral blood mononuclear cells, C.T.L). Then, PBMC was stimulated for 6 days by the addition of an anti-CD3 antibody (BioLegend, Inc., #300438) and an anti-CD28 antibody (BioLegend, Inc., #302934) (final concentration: 0.1 µg/ml each). The amount of CFSE is decreased depending on the number of divisions of cells that have taken up CFSE therein.

Therefore, the ratio of CFSE.sup.low CD4-positive T cells with a signal diluted by proliferation in PBMC was quantified as a CD4-positive T cell proliferation rate by flow cytometry in accordance with a known method (Killer et al., Stem Cell Research & Therapy (2017) 8:100) to evaluate an immunosuppressive effect of the dental pulp-derived cells.

[0264] As a result, the proliferation of CD4-positive T cells induced by simulation with an anti-CD3 antibody and an anti-CD28 antibody was markedly suppressed by the present human dental pulp stem cells (FIG. 11). This demonstrated that the present cells have a strong immunosuppressive effect.

Example 14: Effect on Spinal Cord Injury

[0265] A weight (diameter: 2.5 mm, 10 g) was dropped from a height of 50 mm to the T9-T10 thoracic spinal cord of a rat (Jcl:SD, SPF, CLEA Japan, Inc.) under anesthesia using MASCIS Impactor (Rutgers University, USA) to prepare a spinal cord injury model. 7, 9, 11, and 13 weeks after model preparation, a solvent or the human dental pulp stem cells of the present invention at 1×10^6 cells into the vein and at 5×10^5 cells into the spinal cavity were administered to study motor function-restoring and nerve-regenerating effects on chronic-phase spinal cord injury. Specifically, a hindlimb motor function was evaluated over time by the BBB (Basso-Beattie-Bresnahan) test. 15 weeks after model preparation, the rat was fixed by reflux using 4% paraformaldehyde. Then, the spinal cord including the injury site was collected, and coronal frozen slices were prepared using OCT compound. The slices were stained with H&E or Luxol Fast Blue (LFB), and stained regions were each quantified by image analysis using Image J to evaluate nerve regeneration. Cyclosporin was intraperitoneally administered at a dose of 10 mg/kg every alternate day from 1 day before initial administration.

[0266] As a result, an upward tendency was found in a BBB score from 4 weeks after initial administration of the cells, and statistically significant elevation was found 2 weeks after final administration (FIG. 12A). Both the H&E staining and LFB staining of the spinal cord slices exhibited significant increase in stained area ascribable to the administration of the cells in the epicenter (FIG. 12B). This suggested that the present human dental pulp stem cells are capable of regenerating neural parenchyma and myelin sheath and restoring a motor function upon administration using a clinically implementable route and timing in the chronic phase of spinal cord injury.

Example 15: Effect on Perinatal Hypoxic Ischemic Encephalopathy

[0267] The left carotid artery of a 7-day-old Wistar/ST rat was ligated under anesthesia, and the rat was left for 1 to 2 hours and then left for 1 hour in a hypoxic (8% O.sub.2) environment to prepare a newborn rat hypoxic ischemic encephalopathy model. On the next day, a solvent or the human dental pulp stem cells of the present invention at 1×10^5 cells were administered into the vein to study a motor function improving effect on perinatal hypoxic ischemic encephalopathy. Specifically, a rotarod test was conducted 41 days after administration, and limb motor coordination and endurance were evaluated by measuring a latent time to fall from a rod.

[0268] As a result, a drastically shortened latent time to fall, i.e., an obvious disorder in motor function, was found in the solvent administration group compared with a sham treatment group (normal group), whereas the latent time to fall was rarely shortened in the cell administration group, showing a restored motor function (FIG. 13). This suggested that the present human dental pulp stem cells are capable of recovering symptoms of cerebral palsy in a relatively short period upon administration using a clinically implementable route and timing in perinatal hypoxic ischemic encephalopathy.

Example 16: Effect on Severe Lower Limb Ischemia

[0269] The right common iliac artery and femoral artery and vein of an immunodeficient rat (F344/NJCL/-rnu, CLEA Japan, Inc.) were ligated and cut off to prepare a severe lower limb ischemia model. On the next day, grouping was performed by using the degree of decrease in volume of blood flow compared with normal limbs as an index, and the present human dental pulp

stem cells (2×10^6 cells/head) were then intramuscularly administered to ischemic lower limbs to study an effect on severe lower limb ischemia. A blood flow improving effect based on an angiogenic effect was evaluated on the basis of the volume of blood flow compared with normal limbs. An improving effect on limb necrosis was visually evaluated on the basis of the presence or absence of necrosis up to the heel on an individual basis on Day 7 (n=4 for each group) and Day 14 (n=6 for each group) after model preparation.

[0270] As a result, a sustained decrease in volume of blood flow was found over 14 days after ischemia in a solvent administration group, whereas a markedly restored volume of blood flow was found with each passing day in the cell administration group (FIG. 14A). The cell administration group also exhibited a high limb necrosis-free rate over 14 days and differed statistically significantly from the solvent administration group (FIG. 14B). These results suggested that the present human dental pulp stem cells are capable of restoring blood flow through angiogenesis upon administration using a clinically implementable route and timing in severe lower limb ischemia, and consequently avoiding foot amputation ascribable to limb necrosis.

Example 17: Ability to Produce Cytokine (2)

[0271] In the same manner as in Example 7, the present human dental pulp stem cells were subcultured in 10% hPL/MEM α or 10% FBS/MEM α and then cultured in a basal medium, and the amount of BDNF production and the amount of VEGF production in the resulting culture supernatant were each measured using LEGENDplex™ Human Neuroinflammation panel 1 (BioLegend, Inc., Cat: 740795).

[0272] As a result, the culture supernatant after subculture in 10% hPL/MEM α had a higher concentration ratio to IL-6 and amounts of BDNF and VEGF production per 1×10^6 cells than those of the culture supernatant after subculture in 10% FBS/MEM α (Table 6) (FIG. 15).

TABLE-US-00006

TABLE 6		10% hPL/MEM α	10% FBS/MEM α	Ratio to IL-6	Ratio to IL-6
Donor 1	Donor 2	Donor 3	Donor 4	Mean	Mean
2.19	0.68	1.1		Mean	Mean
				BDNF	0.45
				VEGF	3.34

Example 18: Vascular Endothelial Cell Proliferative Effect (2)

[0273] In the same manner as in Example 10, the culture supernatant of the present human dental pulp stem cells precultured using 10% hPL/MEM α or 10% FBS/MEM α , which was mixed with MEMA at various ratios and further supplemented with 2% (final concentration) FBS were prepared. These cultured supernatants were added to different wells containing Humedia/2% FBS in equal amounts. CCK-8 was added to HUVEC cultured for 3 days after addition of each culture supernatant, and reacted at 37° C. for 4 hours. Then, the number of HUVEC cells was evaluated by measuring OD450.

[0274] As a result, HUVEC supplemented with the culture supernatant of the cells cultured using 10% FBS/MEMO (“FBS”) did not exhibit a proliferative effect as exhibited by HUVEC supplemented with the culture supernatant of the cells cultured using 10% hPL/MEM α (“hPL”) (FIG. 16).

Example 19: Vascular-Like Structure Constructing Effect (2)

[0275] In the same manner as in Example 11, a tube forming effect was studied using HUVEC suspended in the culture supernatant of the present human dental pulp stem cells cultured using 10% FBS/MEM α and HUVEC suspended in the dental pulp stem cell culture supernatant recovered in Example 8.

[0276] As a result, the dental pulp stem cell culture supernatant recovered in Example 8 (“10% hPL/MEM”) was found to promote tube formation as compared with the dental pulp stem cell culture supernatant obtained by culture using FBS (“10% FBS/MEM”) (FIG. 17).

Example 20: Comparison of Ability to Produce Cytokine Among Various Stem Cells

[0277] Human deciduous tooth-derived dental pulp stem cells (SHED, the present human dental pulp stem cells), bone marrow-derived stem cells (BMMSC), and adipose tissue-derived stem cells (ATMSC) were inoculated at 5.76×10^5 cells/well to a 6-well plate, followed

by replacement with Minimum Essential Medium α (MEM α) medium 24 hours after inoculation. The amounts of production of brain-derived neurotrophic factor (BDNF), nerve growth factor (NGF), angiopoietin-like 4 (ANGPTL4), vascular endothelial growth factor (VEGF), stromal cell-derived factor 1 (SDF-1), and monocyte chemoattractant protein-1 (MCP-1) were compared among culture supernatants recovered 48 hours thereafter (FIG. 18; plots in the drawing respectively show results about cells derived from different donors, and the bar represents a mean thereof).

[0278] As a result, the human deciduous tooth-derived dental pulp stem cells were found to have higher ability to produce these liquid factors than that of the other stem cells. It is known that BDNF and NGF are deeply involved in axonal elongation, synaptic transmission, and neuroprotection in the brain nervous system; ANGPTL4 and VEGF are deeply involved in angiogenesis; and SDF1 and MCP-1 were deeply involved as chemokines in cell migration.

Example 21: Effect on Severe Lower Limb Ischemia (2)

[0279] For the severe lower limb ischemia model prepared in Example 16, angiogenesis in a dental pulp stem cell administration group and a solvent administration group was quantitatively evaluated by automatic macro with Image J in tissue staining images using an antibody against a smooth muscle cell marker α SMA (ACTA2).

[0280] As a result, elevation in both of a vessel area and a vessel number in an ischemic lower limb muscle was shown in the cell administration group compared with the solvent administration group (FIG. 19). This demonstrated that the dental pulp stem cells are capable of promoting angiogenesis at a severe ischemic site.

Example 22: Anti-Inflammatory Effect

[0281] A human monocyte leukemia cell line (THP-1) in RPMI1640 (manufactured by Gibco) medium containing 10% FBS and 1 $\times$ penicillin-streptomycin (10% FBS/RPMI1640 medium) was stimulated with 10 ng/ml phorbol 12-myristate 13-acetate (PMA) and inoculated at 5 $\times$ 10⁵ cells/well to a 24-well plate. Transwell (Falcon Culture Insert, 24-well, 0.4 μ m (HD) PET translucent membrane) having a mesh with a pore size of 0.4 μ m, which had been supplemented with Human deciduous tooth-derived dental pulp stem cells (SHED), bone marrow-derived mesenchymal stem cells (BMMSC), adipose tissue-derived mesenchymal stem cells (ATMSC), or permanent tooth-derived dental pulp stem cells (DPSC) at 5 $\times$ 10⁴ cells/well (medium: 10% FBS/RPMI1640 medium) respectively was placed on separate well, and the Culture Insert supplemented with only 10% FBS/RPMI1640 medium was placed on the control wells (Cont). Then, 20 ng/ml IL-4 (M2 macrophage induction conditions) or 1 μ g/ml LPS and 20 ng/ml IFN- γ (M1 macrophage induction conditions) were added thereto. After the resulting cells were cultured for 24 hours, only the THP-1 cells were recovered from each well, and the mRNA levels of transglutaminase-2 (TGM2) and IL-12B were measured by RNA extraction and RT-qPCR.

[0282] As a result, the transcription of TGM2 in the THP-1 cells stimulated for M2 macrophage induction was significantly promoted by coculture with the human deciduous tooth-derived dental pulp stem cells as compared with coculture with the other stem cells (FIG. 20A; plots in the drawing respectively show results about cells derived from different donors, and the bar represents a mean thereof). This demonstrated that the human deciduous tooth-derived dental pulp stem cells are capable of more efficiently promoting the polarization of the THP-1 cells into M2 macrophages having an anti-inflammatory effect than the other stem cells. The transcription of an inflammatory cytokine IL-12B in the THP-1 cells stimulated for M1 macrophage induction was suppressed by coculture with the human deciduous tooth-derived dental pulp stem cells at the same level as in coculture with the other stem cells (FIG. 20B; plots in the drawing respectively show results about cells derived from different donors, and the bar represents a mean thereof). This result demonstrated that the human deciduous tooth-derived dental pulp stem cells act to suppress inflammation.

Example 23: Comparison with Animal-Free Serum-Free Medium

(1) Cell Proliferation Rate

[0283] Human deciduous tooth-derived dental pulp stem cells from four donors were cultured in the same manner as in Example 1. However, culture after cell recovery was performed using 10% hPL/MEMA medium (the medium of the present invention), MesenCult™ ACF Plus Medium Kit (manufactured by STEMCELL Technologies Inc.) (serum-free medium 1), or KBM ADSC-4 (manufactured by Kohjin Bio Co., Ltd.) (serum-free medium 2). The inoculation density was set to from 1.3 to 2.6×10^{sup.3} cells/cm^{sup.2}, and the cells were inoculated depending on the degree of proliferation so as to attain the same timing of passage.

[0284] The numbers of cells in each medium at passages 4, 7, 11, 14, and 16 were measured using a cell counter (NucleoCounter NC-202, manufactured by ChemoMetec) and plotted as a proliferation rate (FIG. 21).

[0285] As a result, the medium of the present invention exhibited a higher cell proliferation rate than that of the other animal-free serum-free media.

(2) Surface Marker Analysis

[0286] Human deciduous tooth-derived dental pulp stem cells from four donors cultured until passage 8 in the same manner as in (1) were subjected to surface marker analysis in the same manner as in Example 2.

TABLE-US-00007 TABLE 7 Donor 9 (positive ratio for surface marker (%)) Serum-free Serum-free Medium of present medium 1 medium 2 invention CD73 99.93 99.95 99.86 CD105 99.80 99.55 99.41 CD90 99.98 99.92 99.92 CD49d 97.85 99.32 99.67 CD51 99.92 99.90 99.94 CD325 1.18 1.07 1.32 CD117 1.26 0.71 1.29 CD31 2.66 12.21 1.79 CD34 2.53 1.49 0.81 CD45 1.24 1.32 0.87

TABLE-US-00008 TABLE 8 Donor 10 (positive ratio for surface marker (%)) Serum-free Serum-free Medium of present medium 1 medium 2 invention CD73 99.95 99.94 99.92 CD105 99.90 98.82 98.82 CD90 99.90 99.97 99.93 CD49d 96.38 98.94 99.90 CD51 99.92 99.91 99.93 CD325 1.34 1.10 0.87 CD117 1.41 0.87 0.80 CD31 8.75 14.03 2.70 CD34 5.47 1.82 1.31 CD45 1.69 0.65 0.61

TABLE-US-00009 TABLE 9 Donor 11 (positive ratio for surface marker (%)) Serum-free Serum-free Medium of present medium 1 medium 2 invention CD73 99.93 99.91 99.93 CD105 99.52 99.59 99.84 CD90 99.97 99.96 99.97 CD49d 78.77 99.49 99.88 CD51 99.94 99.93 99.95 CD325 1.80 1.74 3.09 CD117 1.32 1.24 0.58 CD31 10.83 17.18 1.96 CD34 3.77 2.34 3.39 CD45 0.55 0.08 1.39

TABLE-US-00010 TABLE 10 Donor 12 (positive ratio for surface marker (%)) Serum-free Serum-free Medium of present medium 1 medium 2 invention CD73 99.88 99.94 99.96 CD105 99.67 97.96 99.39 CD90 99.86 99.94 99.92 CD49d 59.38 97.94 99.71 CD51 99.97 99.92 99.92 CD325 1.34 0.74 1.81 CD117 1.56 0.94 0.88 CD31 3.13 6.03 2.28 CD34 4.34 1.84 1.02 CD45 1.21 0.63 0.69

(3) Ability to Produce Cytokine

[0287] Human deciduous tooth-derived dental pulp stem cells from four donors cultured until passage 8 in the same manner as in (1) were evaluated for their ability to produce cytokines in the same manner as in Example 7.

TABLE-US-00011 TABLE 11 Amount of cytokine production (pg/ml) Donor 9 Donor 10 Donor 11 Donor 12 IL-6 Medium of present 29.89833 174.11 240.8367 57.26 invention Serum-free medium 1 19.76167 18.05833 28.4933 16.56833 Serum-free medium 2 7.72 21.22167 68.81833 3.676667 SCF Medium of present 25.04333 29.095 48.93667 24.235 invention Serum-free medium 1 11.08167 9.291667 14.765 10.47167 Serum-free medium 2 9.075 7.22 12.365 7.318333 ANGPTL4 Medium of present 4029.6574 1539.608 1510.591 2690.014 invention Serum-free medium 1 0 174.6353 0 0 Serum-free medium 2 661.47817 279.3081 307.1098 499.6569 BDNF Medium of present 65.87667 60.29 87.36333 95.85 invention Serum-free medium 1 4.936667 3.036667 1.996667 3.438333 Serum-free medium 2 5.555 1.298333 5.313333 4.29 VEGF Medium of present 1248.205 1057.235 491.1667 771.9767 invention Serum-free medium 1 432.4333

182.9317 122.4117 240.2383 Serum-free medium 2 357.3967 423.605 301.2283 198.73
Angiopoietin-2 Medium of present 43.0566 50.79 99.16167 23.13167 invention Serum-free
medium 1 10.135 7.315 12.27667 20.90833 Serum-free medium 2 5.938333 8.49 13.38333
7.638333 β -NGF Medium of present 3.212 8.419833 7.8415 3.85 invention Serum-free medium 1
0.289833 0 0 0 Serum-free medium 2 0 0 0 0 MCP-1 Medium of present 1980 1106.401 3604.247
3420.026 invention Serum-free medium 1 131.519 279.2285 67.07983 253.9732 Serum-free
medium 2 55.032 168.2838 112.3287 194.4633

[0288] In the case of culturing human deciduous tooth-derived dental pulp stem cells in the medium of the present invention, the amounts of production of IL-6, SCF, ANGPTL4, BDNF, VEGF, angiopoietin-2, β -NGF, and MCP-1 were found higher than those in the other serum-free media.

[0289] The ratios of the amounts of production of SCF, ANGPTL4, BDNF, VEGF, angiopoietin-2, and β -NGF to the amount of production of IL-6 in the medium of the present invention were as shown in Table 12.

TABLE-US-00012 TABLE 12 Ratios of amounts of production of SCF, ANGPTL4, BDNF, VEGF, angiopoietin-2, and β -NGF to amount of production of IL-6 Donor 9 Donor 10 Donor 11 Donor 12
SCF 0.84 0.17 0.20 0.42 ANGPTL4 134.78 8.84 6.27 46.98 BDNF 2.20 0.34 0.36 1.67 VEGF
41.74 6.07 2.04 13.48 Angiopoietin-2 1.44 0.29 0.41 0.40 β -NGF 0.11 0.05 0.03 0.07

Example 24: Surface Marker Analysis (2)

[0290] Human deciduous tooth-derived dental pulp stem cells derived from a single donor were chopped and treated with Liberase in the same manner as in Example 1. Then, the tissue was divided into two parts at the stage of cell recovery through a strainer, and the parts were inoculated into culture containers having 10% hPL/ α MEM and 20% FBS/ α MEM, respectively, as a medium, and isolated and cultured. A cell population whose colony detachment was confirmed from the culture container having 10% hPL/ α MEM medium was expansion-cultured in 10% hPL/ α MEM medium, and a cell population whose colony detachment was confirmed from the culture container having 20% FBS/ α MEM medium was expansion-cultured in 10% FBS/ α MEM medium or 20% FBS/ α MEM medium. Each sample was subjected to surface marker analysis by flow cytometry in the same manner as in Example 2 at passage 2. Table 13 shows the surface marker-positive ratios of the cell population expansion-cultured in 10% hPL/ α MEM medium ("10% hPL"), the cell population expansion-cultured in 10% FBS/ α MEM medium ("10% FBS"), and the cell population expansion-cultured in 20% FBS/ α MEM medium ("20% FBS").

TABLE-US-00013 TABLE 13 Positive ratio for surface marker (%) 10% hPL 10% FBS 20% FBS
CD73 99.96 99.95 99.96 CD105 95.68 99.8 98.01 CD90 99.94 99.96 99.96 CD49d 95.31 82.98
79.5 CD51 99.97 100.00 99.97 CD325 0.72 1.06 1.18 CD117 2.28 12.41 5.43 CD31 1.00 1.46 1.17
CD34 0.84 1.13 1.07 CD45 0.92 1.15 1.10

[0291] The amounts of production of SCF and ANGPTL4 per 1×10^6 cells after 48 hours of culture in the medium of the present invention were calculated in the same manner as in Example 7 (Table 14).

TABLE-US-00014 TABLE 14 SCF and ANGPTL4 (ng/dL) produced by human dental pulp-derived cell cultured in 10% hPL/MEM α Donor 9 Donor 10 Donor 11 Donor 12 SCF 0.11 0.13
0.21 0.11 ANGPTL4 17.49 6.68 6.56 11.68

Example 25: Effect on Severe Perinatal Hypoxic Ischemic Encephalopathy

[0292] The right common carotid artery of an SD rat of 7 days after birth was ligated under anesthesia and then cut, and the rat was left for 105 minutes in a hypoxic (8% O₂) environment to prepare a severe hypoxic ischemic encephalopathy model of the newborn rat. At the same time therewith, a sham treatment group was also prepared. At the age of 35 days, grouping was performed by the rotarod method to establish three groups, a sham treatment group, a solvent administration group, and a cell administration group. At the ages of 5 weeks, 7 weeks, 9 weeks, and 11 weeks (four doses at 2-week intervals), the human dental pulp stem cells of the present

invention were administered (1×10^7 cells/kg) into the tail vein. At the age of 13 weeks, a cylinder test was conducted in a blind manner. At the age of 14 weeks, the animals were euthanized, and the whole brain was excised from half the number of animals in each group (5 animals in the sham treatment group and 6 animals each in the solvent administration group and the cell administration group). Then, tissues were fixed in 4% paraformaldehyde-phosphate buffer solution. Cerebrum tissue samples were embedded in paraffine and sliced in accordance with routine methods, and all the samples were stained with HE and immunostained with myelin basic protein (MBP). Since almost all of the individuals in the solvent administration group and the cell administration group lacked a great part of the cerebrum on the disorder side (right side), the total length of MBP was subjected to image analysis in a blind manner for the left cerebrum tissues on the non-disorder side.

[0293] As a result of the cylinder test, the usage rate of the left anterior limb controlled by the brain on the disorder side was significantly high in the cell administration group (“Cell” in the drawing) compared with the solvent administration group (“Solvent” in the drawing). Thus, an improving effect of the administration of the cells on motor dysfunction was found (FIG. 22A).

Immunostaining with MBP was performed for the purpose of studying the relationship between the administration of the cells and histological change in cerebrum tissue, and a nerve fiber length was calculated from a staining image thereof. Significant increase in total nerve fiber length was found in the cell administration group compared with the solvent administration group (FIG. 22B). These results suggest the possibility that under the conditions of this test, the present human dental pulp stem cells elongate (or increase) nerve fiber in the cerebrum on the non-disorder side and contribute to functional restoration of the left anterior limb on the disorder side.

[0294] These results demonstrated that repetitive administration using a clinically implementable administration route and administration timing in severe perinatal hypoxic ischemic encephalopathy improves motor dysfunction of cerebral palsy and nerve elongation on the non-disorder side contributes to a partial mechanism of action thereof.

[0295] As described above, the human deciduous tooth-derived dental pulp stem cells have a high ability to produce cytokines necessary for tissue regeneration such as axonal elongation, synaptic transmission, angiogenesis, and cell migration in the brain nervous system, as compared with other mesenchymal stem cells. On the other hand, the human deciduous tooth-derived dental pulp stem cells have a high ability to suppress the induction of inflammatory macrophages and to induce anti-inflammatory macrophages, and therefore act to suppress inflammation. Hence, the human deciduous tooth-derived dental pulp stem cells are capable of serving as a safer and more highly functional cell medicament.

[0296] Useful cytokines such as SCF, ANGPTL4, BDNF, VEGF, angiopoietin-2, β -NGF, and MCP-1 can be highly produced while the production of an inflammatory cytokine (IL-6) is suppressed, by culturing the human deciduous tooth-derived dental pulp stem cells in a serum-free medium containing no FBS in the presence of hPL, as compared with culture in a medium containing FBS.

INDUSTRIAL APPLICABILITY

[0297] The dental pulp stem cells of the present invention have multipotency, a neural progenitor cell proliferative effect, a neurite elongating effect, a vascular endothelial cell proliferative effect, a vascular endothelial cell recruiting effect, a vascular-like structure constructing effect, and an immunosuppressive effect and are useful in the treatment of spinal cord injury, ischemic disease, inflammatory disease, neurodegenerative disease, and the like.

Test Example 1. Preparation of Human Deciduous Dental Pulp Stem Cell

[0298] Deciduous dental pulp stem cells were prepared from a human deciduous tooth that fell out. Specifically, the preparation was performed as follows: dental pulp was scraped out of the human deciduous tooth, chopped, and then treated with Liberase (Roche) added at a final concentration of 0.05 mg/ml with stirring at 37° C. for 15 minutes. A reaction supernatant containing cells was

recovered, and the remaining dental pulp tissue was further treated with Liberase added at 0.05 mg/ml with stirring at 37° C. for 15 minutes. Cells were recovered from the treated dental pulp tissue through a 70 mm strainer, combined with the supernatant recovered first, and centrifuged. The recovered cells were inoculated to a culture container using MEMX (Gibco) containing 10% hPL, and isolated and cultured for 9 days or longer until colonies were formed, grew, and were detached. Medium replacement was performed 2 days and 9 days after inoculation. After colony detachment was confirmed, cells were recovered using TrypLE Select (Gibco) and expansion-cultured using the same medium as above.

Test Example 2. Evaluation of Therapeutic Effect of Deciduous Dental Pulp Stem Cell on Cerebral Palsy

Test Example 2-1. Preparation of Cerebral Palsy Model Rat

[0299] HIE models (Rice-Vannucci models) were prepared. Specifically, left common carotid artery ligation and hypoxic loading (O.sub.2 8% for 60 minutes) were performed for 7-day-old male Wistar/ST rats. At the same time therewith, rats of a sham operation group (Sham group) were also prepared.

Test Example 2-2. Grouping by Behavioral Test

[0300] The model rats prepared in Test Example 2-1 were subjected to a horizontal ladder test at the age of 25 to 28 days. Specifically, each rat was allowed to walk on a horizontal ladder, and the gait of a paralyzed leg was scored with 1 to 4 points (0: normal walking, 1: the paralyzed leg slips on the ladder but does not fall out, 2: the paralyzed leg falls out of the ladder, though the next step is normal, 3: the paralyzed leg falls out of the ladder, and the next step is not normal, 4: both legs fall out of the ladder). A score per step (score/step) was calculated. Only individuals given significantly abnormal findings as compared with the Sham group were selected. The selected model rats were divided into two groups (cell administration (Cell) group and a vehicle (Vehicle) group) such that the scores of the test were at the same level between the groups.

Test Example 2-3. Administration of Cell

[0301] For the Cell group, 1×10^6 cells of the deciduous dental pulp stem cells (Test Example 1) were administered to the tail vein at the age of 35 days (single-dose administration). In the test of another batch, for the Cell group, 1×10^6 cells of the deciduous dental pulp stem cells (Test Example 1) were administered to the tail vein at the ages of 35 days, 49 days, and 63 days (multiple-dose administration). On the other hand, for the Vehicle group, only a medium containing no deciduous dental pulp stem cells was administered in the same manner as in the Cell group.

Test Example 2-4. Evaluation Test

[0302] The Cell group, the Vehicle group, and the Sham group were subjected to a horizontal ladder test, a cylinder test, and shuttle avoidance.

[0303] The horizontal ladder test was conducted by the same method as in Test Example 2-2.

[0304] In the cylinder test, each rat was placed in a cylinder. Anterior limb [left (healthy side), right (affected side), or both sides] contacts of the rat to the wall were recorded to evaluate a use rate of the limb on the affected side. Scores were calculated according to the expression: $(L-R)/(L+R+B)$ (L: left, R: right, B: both). Stronger damage decreases the rate of use of the right (affected side) limb (elevates the score).

[0305] In the shuttle avoidance, each rat was placed in a cage, warned by light and sound, and electrically stimulated if there was no avoidance. This operation was repeated to evaluate memory learning capability. 10×6 sets per day (a total of 60 sets) were tested for 2 consecutive days, and an avoidance rate for each set was evaluated.

[0306] The results of the horizontal ladder test are shown in FIGS. 23 to 25. The administration of the deciduous dental pulp stem cells was found to improve a motor function evaluated by the horizontal ladder test. The multiple-dose administration tended to exert a stronger improving effect.

[0307] The results of the cylinder test are shown in FIG. 26. The administration of the deciduous dental pulp stem cells was found to improve a motor function evaluated by the cylinder test. The

multiple-dose administration exhibited a more marked improving effect than that of the single-dose administration.

[0308] The results of the shuttle avoidance are shown in FIG. 27. The administration of the deciduous dental pulp stem cells was found to improve a memory learning function evaluated by the shuttle avoidance. The multiple-dose administration exhibited a more marked improving effect than that of the single-dose administration.

Test Example 3. Measurement of Body Weight and Brain Weight Ratio

[0309] Body weights and brain weight ratios (cerebrum weight/cerebellum+brain stem weight) at the completion of the test (the age of 110 days) were measured for each group of Test Example 2.

[0310] The results of the body weight measurement are shown in FIG. 28, and the results of the brain weight ratio measurement are shown in FIG. 29. The body weight exhibited a significantly high value in the Sham group; however, it did not differ between the Cell group and the Vehicle group. The brain weight ratio (cerebrum weight/cerebellum+brain stem weight) was a significantly low value in the Vehicle group.

Test Example 4. Mechanism Analysis by Comprehensive Evaluation of Target Protein

[0311] The deciduous dental pulp stem cells were administered in multiple doses in the same manner as in the multiple-dose administration of Test Example 2. The brain was homogenized 2 weeks after final administration of the cells, and extracted proteins were comprehensively analyzed by liquid chromatography-mass spectrometry (LC/MS). Of all the analyzed proteins (n=1696), proteins detected at a level below detection sensitivity in 5 or more out of six samples in the Vehicle group or the Cell group were excluded (n=204). 61 types of proteins found to vary significantly in Vehicle group vs Cell group were obtained (p-value<0.05 (student's t-test) FDR<0.1 (Storey method)). These are a group of proteins presumably influenced by the administration of the deciduous dental pulp stem cells except for the influence of HIE. As a result of extracting clusters with an enrichment score>2 by GO analysis using DAVID, proteins involved in neurogenesis, morphological formation of cells, axonal new formation, and phosphorylation processes were enriched.

Test Example 5. In Vivo Kinetic Evaluation of Deciduous Dental Pulp Stem Cell Using Quantum Dot

[0312] Quantum dot-labeled deciduous dental pulp stem cells (1×10^6 cells/body) were administered to the tail vein of each rat. Evaluation was conducted at four time points (A: 1 hour after injection, B: 1 day, C: 2 days, D: 1 week) after intravenous injection of the cells. The target organ (brain) was taken out at each time point and evaluated for fluorescence intensity using an in vivo imaging system. Fluorescence intensity ratio=fluorescence intensity of each individual in the Cell group/mean fluorescence intensity of the Vehicle group at each time point was calculated on the basis of the fluorescence intensity. The results are shown in FIG. 30.

Test Example 6. Evaluation of Neurogenetic Effect Using Deciduous Dental Pulp Stem Cells

Test Example 6-1: Neurogenesis Marker

[0313] The deciduous dental pulp stem cells were administered in multiple doses to each HIE model (Rice-Vannucci model) rat in the same manner as in Test Examples 2-1 to 2-3.

Bromodeoxyuridine (BrdU) at 50 mg/kg was intraperitoneally administered once a day for a total of 3 days from the day following the third administration of the cells.

[0314] At 2 weeks after the third administration of the cells, the rat was perfusion-fixed in paraformaldehyde (PFA), and a brain tissue was collected and embedded in a paraffin block. This block was sliced to prepare brain tissue slices, followed by (1) immunohistological staining using an anti-BrdU antibody and (2) fluorescent double immunostaining with an anti-BrdU antibody and an anti-doublecortin (DCX) antibody.

[0315] The hippocampal dentate gyrus and the subventricular zone on the damage side were evaluated for each slice, and the number of BrdU-positive cells and the ratio of BrdU/DCX double stain-positive cells to the BrdU-positive cells were calculated.

[0316] As a result, in the immunohistological staining with an anti-BrdU antibody, BrdU-positive cells were found in the innermost layer of the hippocampal dentate gyrus on the damage side, the subependymal layer of the lateral ventricle, and the corpus striatum in the Cell group. In the fluorescent double staining, the ratio of BrdU/DCX double stain-positive cells in the hippocampal dentate gyrus on the damage side was significantly increased in the Cell group (FIG. 31). Further, the number of BrdU-positive cells was significantly increased in the Cell group (FIG. 32).

Test Example 6-2: Neural Marker

[0317] Each individual (Vehicle, Cell, and Sham groups) subjected to the behavioral test in Test Example 2 was perfusion-fixed in PFA at the age of 4 months after birth, and a brain tissue was collected and embedded in a paraffin block. This block was sliced to prepare brain tissue slices, followed by immunohistological staining using an anti-NeuN antibody. The number of NeuN-positive cells in the hippocampal dentate gyrus on the damage side was measured in these slices by the stereology method.

[0318] As a result, the number of NeuN-positive cells in the hippocampal dentate gyrus exhibited a significantly high value in the Cell group compared to the Vehicle group (FIG. 33).

[0319] The results of this test demonstrated that intravenously administered dental pulp stem cells are capable of migrating into the brain of a recipient. The results also demonstrated that the administration of the dental pulp stem cells promotes neurogenesis in the damaged brain, particularly, the hippocampal dentate gyrus.

[0320] The test results disclosed in the present invention suggest that neurogenesis is involved in improvement in motor function or memory learning found upon administration of the deciduous dental pulp stem cells to an individual with cerebral palsy.

Test Example 7. Effect on Perinatal Hypoxic-Ischemic Encephalopathy

Test Example 7-1: Preparation of Dental Pulp Stem Cell

[0321] In a sterile environment, dental pulp was scraped out of an extracted tooth (human deciduous tooth) within 48 hours after tooth extraction, chopped, and then treated with Liberase (MNP-S GMP Grade, Roche) added at a final concentration of 0.05 mg/ml with stirring at 37° C. for 15 minutes in the same manner as in Test Examples 1. A reaction supernatant containing cells was recovered, and the remaining dental pulp tissue was further treated with Liberase added at 0.05 mg/ml with stirring at 37° C. for 15 minutes. Cells were recovered from the treated dental pulp tissue through a 70 mm strainer, combined with the supernatant recovered first, and centrifuged. The recovered cells were inoculated to a culture container (CellBind T75 flask) using MEM& (Gibco) containing 10% hPL (AventaCell BioMedical Co., Ltd.), and isolated and cultured for 9 days or longer until colonies were formed, grew, and were detached. Medium replacement was performed 2 days and 9 days after inoculation. After colony detachment was confirmed, cells were recovered using TrypLE Select (Gibco) and expansion-cultured using the same medium as above.

Test Example 7-2: Evaluation Test in Cerebral Palsy Model Rat

[0322] The left carotid artery of a 7-day-old Wistar/ST rat was ligated under anesthesia, and the rat was left for 1 to 2 hours and then left for 1 hour in a hypoxic (8% O₂) environment to prepare a newborn rat hypoxic ischemic encephalopathy model. On the next day, a solvent or the human dental pulp stem cells of the present invention at 1×10^5 cells were administered into the vein to study a motor function improving effect on perinatal hypoxic ischemic encephalopathy. Specifically, a rotarod test was conducted 41 days after administration, and limb motor coordination and endurance were evaluated by measuring a latent time to fall from a rod.

[0323] As a result, a drastically shortened latent time to fall, i.e., an obvious disorder in motor function, was found in the solvent administration group (Vehicle) compared with a sham treatment group (normal group, Sham), whereas the latent time to fall was rarely shortened in the cell administration group (Cell), showing a restored motor function (FIG. 34). This suggested that the present human dental pulp stem cells are capable of recovering symptoms of cerebral palsy in a relatively short period upon administration using a clinically implementable route and timing in

perinatal hypoxic ischemic encephalopathy.

Test Example 8. Effect on Severe Perinatal Hypoxic Ischemic Encephalopathy

[0324] The right common carotid artery of an SD rat of 7 days after birth was ligated under anesthesia and then cut, and the rat was left for 105 minutes in a hypoxic (8% O₂) environment to prepare a severe hypoxic ischemic encephalopathy model of the newborn rat. At the same time therewith, a sham treatment group was also prepared. At the age of 35 days, grouping was performed by the rotarod method to establish three groups, a sham treatment group (Sham), a solvent administration group (Vehicle), and a cell administration group (Cell). At the ages of 5 weeks, 7 weeks, 9 weeks, and 11 weeks (four doses at 2-week intervals), the human dental pulp stem cells of the present invention were administered (1×10^7 cells/kg) into the tail vein. At the age of 13 weeks, a cylinder test was conducted in a blind manner. At the age of 14 weeks, the animals were euthanized, and the whole brain was excised from half the number of animals in each group (5 animals in the sham treatment group and 6 animals each in the solvent administration group and the cell administration group). Then, tissues were fixed in 4% paraformaldehyde-phosphate buffer solution. Cerebrum tissue samples were embedded in paraffine and sliced in accordance with routine methods, and all the samples were stained with HE and immunostained with myelin basic protein (MBP). Since almost all of the individuals in the solvent administration group and the cell administration group lacked a great part of the cerebrum on the disorder side (right side), the total length of MBP was subjected to image analysis in a blind manner for the left cerebrum tissues on the non-disorder side.

[0325] As a result of the cylinder test, the usage rate of the left anterior limb controlled by the brain on the disorder side exhibited a significantly high value in the cell administration group compared with the solvent administration group. Thus, an improving effect of the administration of the cells on motor dysfunction was found (FIG. 35A). Immunostaining with MBP was performed for the purpose of studying the relationship between the administration of the cells and histological change in cerebrum tissue, and a nerve fiber length was calculated from a staining image thereof.

Significant increase in total nerve fiber length was found in the cell administration group compared with the solvent administration group (FIG. 35B). These results suggest the possibility that under the conditions of this test, the present human dental pulp stem cells elongate (or increase) nerve fiber in the cerebrum on the non-disorder side and contribute to functional restoration of the left anterior limb on the disorder side.

[0326] These results demonstrated that repetitive administration using a clinically implementable administration route and administration timing in severe perinatal hypoxic ischemic encephalopathy improves motor dysfunction of cerebral palsy and nerve elongation on the non-disorder side contributes to a partial mechanism of action thereof.

Test Example 9. Effect on Neurogenesis

Test Example 9-1: Coculture Experiment with Neural Stem Cell

[0327] Neural stem cells (NSC) (Sigma-Aldrich Co., LLC, SCR022) recovered from the hippocampal dentate gyrus of an adult rat were primarily cultured (medium: 100 ml of DMEM/F-12 (Dulbecco's Modified Eagle Medium/Nutrient Mixture F-12), 2.0 ml of B-27 Supplement (50×) serum free, 1.0 ml of GlutaMAX Supplement, and 20 μl of basic fibroblast growth factor (0.1 mg/ml)) in a 24-well plate in accordance with the attached protocol and used in subsequent experiments.

[0328] After a lapse of 24 hours from the primary culture of NSC, human deciduous tooth-derived dental pulp stem cells (SHED) obtained in the same manner as in Test Example 1, human bone marrow-derived mesenchymal stem cells (BM-MSC), or human skin fibroblasts inoculated (2×10^4 cells/well for each cell type) to an insert (Corning Inc., Culture Insert, 24-well, 0.4 μm PET transparent, part number: 353095) were placed on each well inoculated with NSC (1×10^4 cells/well), and cocultured in a serum-free medium (Gibco, MEM & nucleosides no phenol red INV, part number: 41061029). NSC cultured alone was

used as a coculture control.

Test Example 9-2: Cell Staining

[0329] At 48 hours after the start of coculture, a cell supernatant was recovered, and the cells on the plate were fixed in PFA. These cells were stained with Hoechst (nucleus), an anti-SOX2 antibody (neuron marker), and an anti-S-100 antibody (astrocyte marker). As a result of the staining, the cells were SOX2-positive; however, were rarely S-100-positive. Thus, SHED in the culture for 48 hours was confirmed to have no influence on the induction of differentiation of NSC.

Test Example 9-3: Evaluation of Cell Proliferation Rate

[0330] Among the cells stained in Test Example 9-2, SOX2-positive cells were counted by the stereology method and normalized with the number of control cells to calculate a cell proliferation rate (growth rate).

[0331] As a result, SHED was found to exhibit the highest proliferative effect on NSC (FIG. 36).

Test Example 9-4: Evaluation of BDNF Concentration in Culture Supernatant

[0332] For the coculture performed in Test Example 9-3, the concentration of brain-derived nutritional factor (BDNF) in the culture supernatant on the NSC culture side recovered before cell fixation was measured by ELISA. As a result, the coculture with SHED was found to have a significantly high BDNF concentration in the culture supernatant (FIG. 37).

[0333] These results indicated that SHED contributes to the proliferation of NSC by coculture with the NCS, and demonstrated that its effect was greater than that of BM-MSC or fibroblasts.

[0334] In addition, the culture supernatant of NSC cocultured with SHED was found to be rich in BDNF.

[0335] The results of this test indicate the possibility that SHED contributes to neurogenesis by promoting the proliferation of NSC remaining at a degeneration site of a brain nerve tissue, for example, through the production of BDNF.

[0336] All publications, patents, and patent applications cited herein are incorporated herein by reference in their entirety.

Claims

1. A stem cell population derived from human dental pulp, wherein 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive, and wherein the human dental pulp is human deciduous dental pulp.
2. The stem cell population according to claim 1, wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a serum-free medium containing a human platelet lysate.
3. The stem cell population according to claim 1, wherein 90% or more of the stem cell population is CD325-negative or CD51-positive.
4. The stem cell population according to claim 1, wherein the stem cell population has at least one of the following characteristics 1) to 3): 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.
5. The stem cell population according to claim 1, wherein the stem cell population has at least one of the following characteristics 1) to 3): 1) producing at least 0.1 ng of SCF (stem cell factor) in 48 hours per 1×10^6 cells, 2) producing at least 500 ng of BIGH3 (transforming growth factor-beta-induced) in 48 hours per 1×10^6 cells, 3) producing at least 5 ng of ANGPTL4 in 48 hours per 1×10^6 cells.
6. The stem cell population according to claim 1, wherein 90% or more of the stem cell population is CD117-negative, CD73-positive, CD90-positive, and CD105-positive; and the stem cell population has an ability to produce any of BDNF, NGF, ANGPTL4, SDF-1 or MCP-1 at two times

or more higher than bone marrow-derived mesenchymal stem cells (BMMSC), or adipose tissue-derived mesenchymal stem cells (ATMSC).

7. The stem cell population according to claim 1, wherein the amount of production of IL-6 is reduced compared to a stem cell population derived from human deciduous dental pulp prepared using a medium containing FBS.

8. The stem cell population according to claim 1, wherein the amount of production of one or more cytokines selected from the group consisting of SCF, ANGPTL4, BIGH3, BDNF, NGF, VEGF, SDF-1, MCP-1, and angiopoietin-2 is enriched compared to a stem cell population derived from human deciduous dental pulp prepared using a medium containing FBS.

9. The stem cell population according to claim 1, wherein the stem cell population is obtained by the method comprising the steps of: enzymatically isolating cells from the human dental pulp; culturing the cells in a serum-free medium containing a human platelet lysate; and obtaining a colony-forming cell population as the stem cell population.

10. The stem cell population according to claim 9, wherein the method further comprises the step of expanding the obtained stem cell population for 2 to 8 passages for expansion.

11. A culture supernatant of the stem cell population according to claim 1, wherein the culture supernatant has at least one of the following characteristics 1) to 3): 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

12. A composition comprising the stem cell population according to claim 1 and a pharmaceutically acceptable carrier.

13. The composition according to claim 12, wherein the stem cell population is obtained by culturing cells enzymatically isolated from the human dental pulp in a serum-free medium containing a human platelet lysate.

14. The composition according to claim 12, wherein the stem cell population has at least one of the following characteristics 1) to 3): 1) producing SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, 2) producing ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, 3) producing BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

15. The composition according to claim 12, wherein the composition has at least one of the following characteristics 1) to 3): 1) comprising SCF (stem cell factor) at a weight ratio of 1/10 or more relative to IL-6, 2) comprising ANGPTL4 (angiopoietin-like 4) at a weight ratio of 5 or more relative to IL-6, 3) comprising BIGH3 (transforming growth factor-beta-induced) at a weight ratio of 450 or more relative to IL-6.

16. A method for producing a stem cell population derived from human dental pulp, the method comprising the steps of: enzymatically isolating cells from the human dental pulp; culturing the cells in a serum-free medium in the presence of a human platelet lysate; and obtaining a colony-forming cell population as the stem cell population, wherein the human dental pulp is human deciduous dental pulp.

17. The method according to claim 16, wherein the medium is animal-free.

18. The method according to claim 16, wherein 90% or more of the resulting stem cell population is CD117-negative.

19. The method according to claim 16, further comprising the step of expanding the obtained stem cell population for 2 to 8 passages for expansion.
